JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 100, NO. B4, PAGES 6073–6091, APRIL 10, 1995

# Paleomagnetism of the Upper Cretaceous strata of Mount Tatlow: Evidence for 3000 km of northward displacement of the eastern Coast Belt, British Columbia

P. J. Wynne and E. Irving

Pacific Geoscience Centre, Geological Survey of Canada, Sidney, British Columbia, Canada

J. A. Maxson[1] and K. L. Kleinspehn

Department of Geology and Geophysics, University of Minnesota, Minneapolis

**Abstract.** This study contributes the first paleomagnetic data from dated Cretaceous units of the Coast Belt (51.3°N, 123.8°W) in which paleohorizontal bedding surfaces can be identified with certainty. We sampled 38 sites through a 1.5-km stratigraphic interval of Cenomanian Silverquick sedimentary strata and Cenomanian to Campanian Powell Creek volcanics in the Tatlow syncline in the Tyaughton-Methow Basin. Magnetizations at 21 sites have high unblocking temperatures ($T_{UB}$) and pass the tilt test (precision $k = 5$ before untilting, maximum $k = 48$ after 100% untilting). Magnetic minerals of volcanics show original textures, mean declinations in volcanic and sedimentary rocks are statistically identical, and polarity is consistently normal. The mean inclination (55.4° ± 3.4°, probability $P = 0.05$) of this pretilting magnetization is 20.1° ± 4.0° shallower than that expected if the rocks were deposited in their present position relative to cratonic North America. Field and laboratory observations rule out other causes for these anomalous shallow inclinations. These rocks therefore have been displaced northward by 3000 ± 500 km relative to cratonic North America since mid-Cretaceous time. This displacement is almost 3 times greater than that estimated from mid-Cretaceous volcanics in the now-adjacent Intermontane Belt. Evidently the two belts were not amalgamated until latest Cretaceous or Early Tertiary time. These results are inconsistent with all previous geologically based tectonic reconstructions of the Cordillera. Magnetizations at eight sites in Silverquick and Powell Creek beds have low $T_{UB}$ and directional characteristics similar to those in three younger diorite intrusions. Reversed and normal polarities occur, the rocks are commonly highly altered, and maximum precision occurs after 40% untilting. The magnetizations therefore are interpreted as overprints of latest Cretaceous age acquired during folding.

## Introduction

Paleomagnetic studies of several mid-Cretaceous plutons and other rocks from the Coast Belt of British Columbia and northern Washington (Figure 1) have yielded magnetization directions with inclinations shallower than those calculated from coeval rocks of the North American craton. Some workers have proposed that these discordances imply ≈3000 km of northward orogen-parallel displacement of the Coast-Cascade Belt relative to the craton [*Beck and Noson*, 1972; *Beck*, 1976, 1980, 1991; *Irving*, 1979; *Beck et al.*, 1981; *Irving et al.*, 1985; *Bogue et al.*, 1989]. Others have argued that the discordances are largely the result of tilting with subordinate orogen-parallel translation [*Symons*, 1974, 1977; *Butler et al.*, 1989; *Vandall and Palmer*, 1990; *Vandall*, 1993].

This dilemma can be resolved by (1) estimating the tilt of plutons from petrologic depth indicators thus allowing the magnetization direction to be corrected to the paleohorizontal plane, and (2) by obtaining observations from Cretaceous stratified rocks in which accurate bedding attitudes can be determined. The first test has already been carried out for the 96-93 Ma Mount Stuart batholith (Figure 1); *Ague and Brandon* [1992], on the basis of hornblende geobarometry, have corrected the paleomagnetic observations [*Beck et al.*, 1981] for tilt and obtained an estimate of motion from the south of 2900 ± 700 km. We describe the first paleomagnetic results from Cretaceous bedded rocks of the Coast Belt, which allow us to carry out the second test.

The sedimentary and volcanic sequence studied is particularly suitable for six reasons. They contain thinly bedded, planar-laminated siltstones and fine-grained sandstones of fluvial overbank origin from which paleohorizontal surfaces can be determined with precision. The $^{40}Ar/^{39}Ar$ data provide the age of the Powell Creek volcanics. The strata occur in a syncline, the Tatlow syncline (Figure 2), with an interlimb

[1]Now at Department of Geology, Carleton College, Northfield, Minnesota.

Copyright 1995 by the American Geophysical Union.

Paper number 94JB02643.
0148-0227/95/94JB-02643$05.00

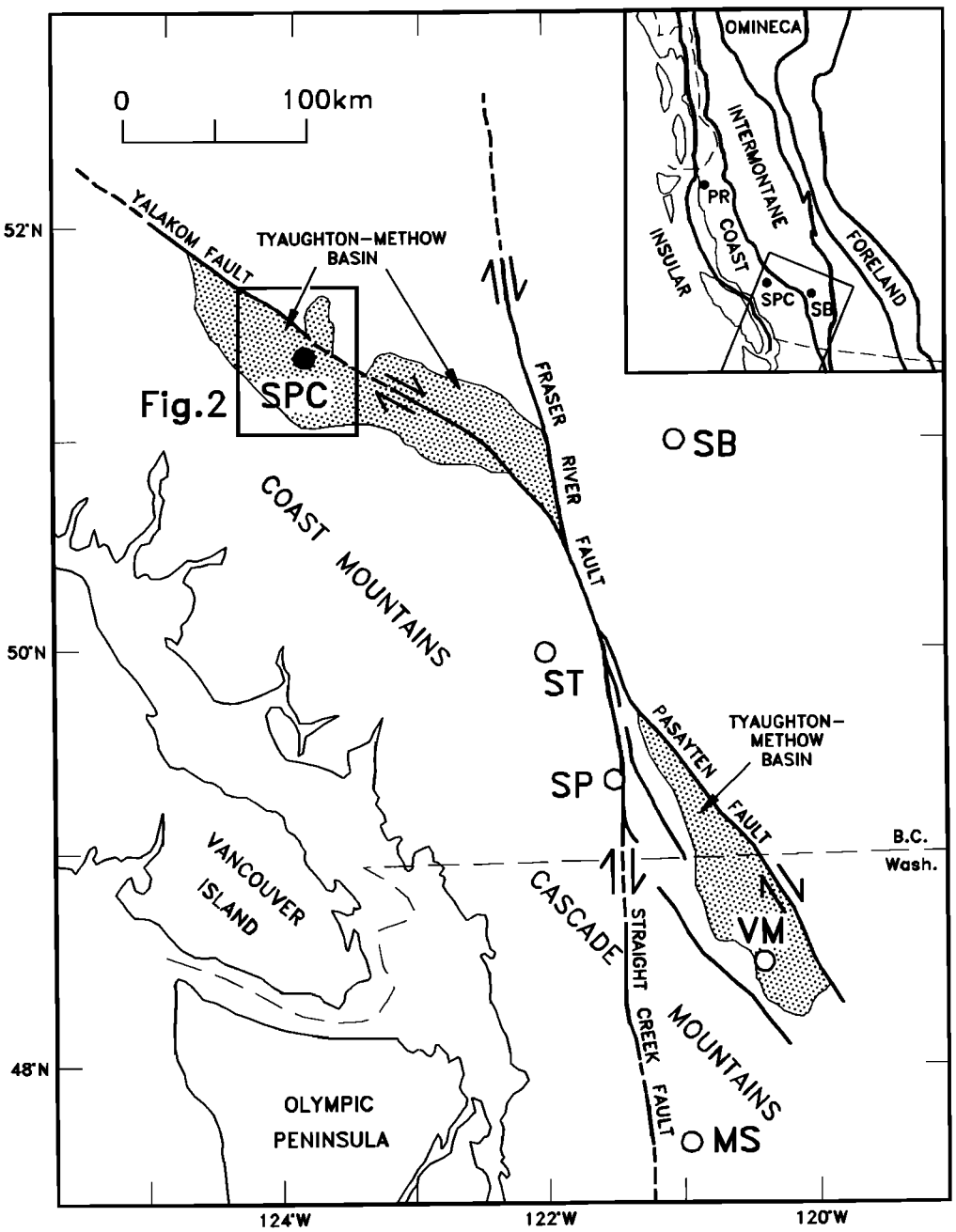



**Figure 1.** Regional geology of the study area. From *Kleinspehn* [1985] and *Wheeler and McFeely* [1991]. Paleomagnetic studies of Cretaceous rocks discussed in the text: MS, Mount Stuart batholith; PR, plutons of the Prince Rupert area; SB, Spences Bridge Group; SP, Spuzzum pluton; SPC, Silverquick conglomerate and Powell Creek volcanics; ST, Spetch Creek pluton; VM, Ventura Member of the Midnight Peak Formation. Morphogeologic belts in inset.

angle of 40°, allowing a rigorous fold test to be made. The >1-km-thick sequence represents a time interval sufficient to average out secular variation of the Earth's magnetic field. Although local contact metamorphism occurred beneath the Miocene Chilcotin Basalt (Figure 2) that overlies unconformably Silverquick strata, regional metamorphic grade is no higher than zeolite facies. Finally, the sequence contains extrusive and intrusive rocks and different types of sedimentary rock, making it possible to compare magnetizations acquired by different processes and to show thereby that the magnetizations accurately record the paleofield.

## Geology

The Silverquick conglomerate (informal nomenclature of *Glover et al.* [1988] and *Garver et al.* [1989], equivalent to Kingsvale Group sediments of *Jeletzky and Tipper* [1968]) in the study area is predominantly fluvial, consisting of channelized sandstone and conglomerate, and planar-laminated overbank deposits of siltstone and mudstone. Strata in the study area are significantly finer grained than in the southeastern Tyaughton basin, where the unit was originally designated as a conglomerate.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

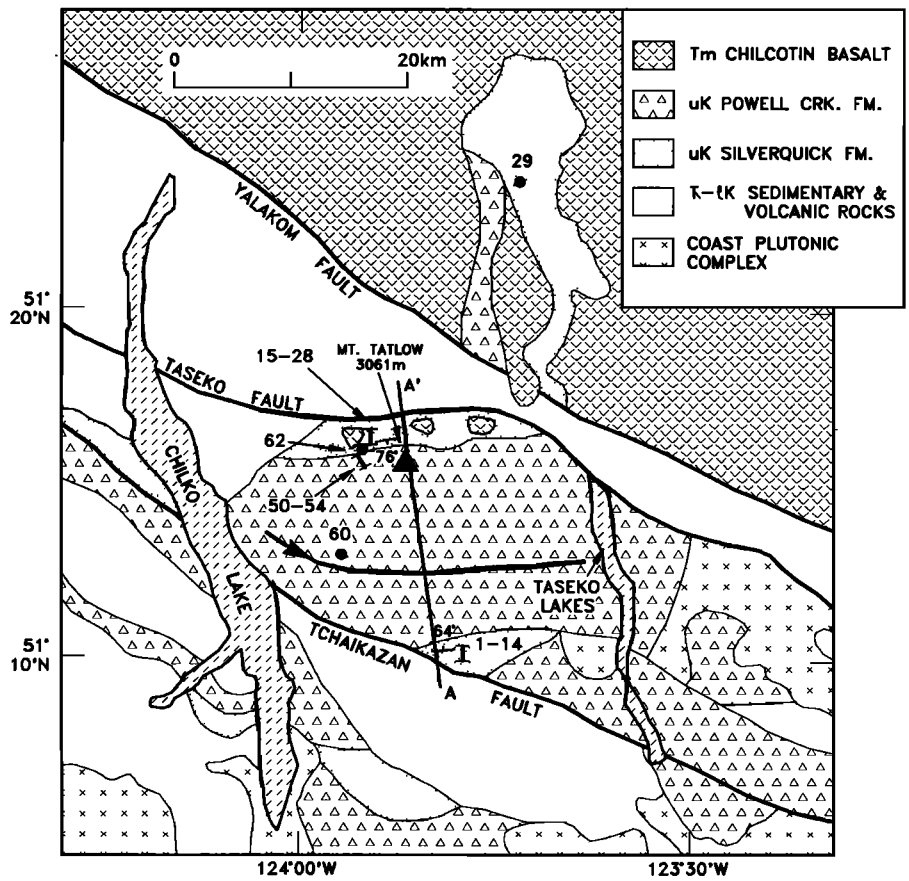



**Figure 2a.** Geological map of Tatlow syncline and adjacent area. The Tatlow syncline, traces of major Late Cretaceous-Early Tertiary dextral faults, and Miocene Chilcotin basalt are shown. Sample sites 1-14 (south limb), 15-28 (north limb, section 1), and 50-54 (north limb, section 2) and outlying sites 29, 60, and 62 are shown. Map modified from *Tipper* [1969, 1978].

Hematite and carbonate cements are common, especially in spheroidal concretions (10 to 50 cm) in siltstone and sandstone. Paleosols occur with increasing frequency upward in the section. The upper beds of the sampled sequence contain alternating red and green mottled horizons with abundant caliche nodules, indicating that primary cementation was partially pedogenic. About one half of the samples are very fine to medium-grained lithic arenites and siltstones (Table 1). Thinly bedded, planar-laminated overbank deposits were targeted because they reliably record paleohorizontal planes. A subordinate set of samples from ripple cross-laminated overbank strata also was taken to test the influence of bedforms. The remainder of Silverquick samples is from concretions in channel and overbank strata. Bedding is draped around the concretions indicating that they were lithified early in diagenesis, before compaction and lithification of the surrounding sediments. Twenty-seven horizons were sampled, spanning 1 km stratigraphically.

The Powell Creek volcanics (informal nomenclature of *Glover et al.* [1988], equivalent to Kingsvale Group volcanics of *Jeletzky and Tipper* [1968]) overlie the Silverquick conglomerate and consist of subaerial andesitic to basaltic lavas beds, breccias and lahar deposits with fluvial interbeds. Each volcanic flow is 1 to 3 m thick. Four massive lavas and one breccia unit were sampled (Table 1) spanning a stratigraphic thickness of approximately 200 m. Bedding

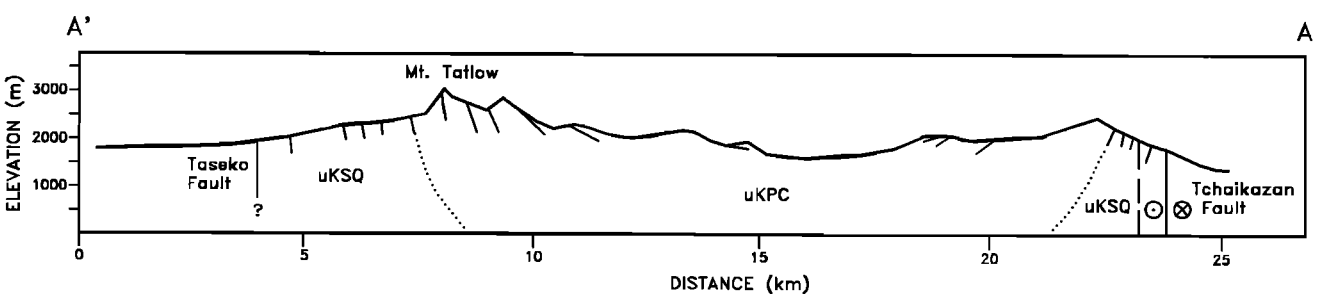



**Figure 2b.** Cross section of the Tatlow syncline, showing inferred subsurface contact between Silverquick (uKSQ) and Powell Creek (uKPC) strata, the Tchaikazan Fault, and the Taseko Fault.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 1.** Results by Site

| Site | Rock Type | *h* | *c* | *s* | *BDDA*, deg | *BD*, deg | *M*, ($10^{-2}$ A/m) | $D_H$, deg | $I_H$, deg | *k* | $\alpha_{95}$, deg |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | *Tatlow Syncline, South Limb (51.3°N, 123.8°W)* | | | | | | |
| 1 | fg cross-lam sandstone (S) | | 5 | 6 | 355 | 66 | 0.38 | 244 | 48 | 19 | 16† |
| 2 | vfg pl lam sandstone (S) | | 6 | 9 | 354 | 52 | 0.69 | 226 | 61 | 76 | 6 |
| 3 | pl lam siltstone (S) | | 9 | 16 | 356 | 66 | 0.71 | 223 | 61 | 55 | 5 |
| 4 | siltstone concretions (S) | | 6 | 11 | 352 | 63 | 3.11 | 202 | 46 | 204 | 3 |
| 5 | sandstone concretions (S) | | 7 | 14 | 357 | 60 | 0.63 | 207 | 47 | 92 | 4 |
| 6 | siltstone concretions (S) | 3 | 8 | 12 | 006 | 67 | 0.50 | 206 | 51 | 119 | 4 |
| 7 | fg cross-lam sandstone (S) | | 6 | 10 | 352 | 65 | 1.22 | 212 | 42 | 104 | 5 |
| 8 | pl lam siltstone (S) | 2 | 9 | 17 | 006 | 64 | 1.07 | 203 | 41 | 43 | 5 |
| R5 | sandstone (concretions)(S) | 7 | 9 | 9 | 355 | 65 | 0.02 | 205 | 50 | 39 | 8 |
| 9 | mg cross-lam sandstone (S) | | 6 | 12 | 006 | 76 | 1.32 | 195 | 52 | 88 | 5 |
| 10 | diorite sill | | 6 | 10 | 006 | 76 | 9.29 | 211 | 64 | 37 | 8 |
| 11* | sandstone concretions (S) | | 6 | 10 | 352 | 90 | 0.34 | 210 | 59 | 135 | 4 |
| 12* | mg cross-lam sandstone (S) | 3 | 5 | 6 | 348 | 92 | 0.99 | 209 | 66 | 240 | 4 |
| 13* | porphyritic diorite stock | 2 | 6 | 9 | 348 | 92 | 0.71 | 144 | 53 | 359 | 3 |
| 14* | basaltic breccia (P) | 4 | 5 | 8 | 348 | 92 | 24.70 | 167 | 64 | 25 | 11 |
| | | | | | *Tatlow Syncline, Section 1 of North Limb (51.4°N, 123.9°W)* | | | | | | |
| 28 | pl lam siltstone (S) | 2 | 4 | 6 | 179 | 76 | 0.69 | 331 | -17 | 10 | 22† |
| 27* | sandstone concretions (S) | 3 | 9 | 12 | 170 | 76 | 0.07 | 352 | 57 | 89 | 5 |
| 26* | sandstone concretions (S) | 3 | 5 | 9 | 170 | 76 | 0.07 | 345 | 55 | 76 | 6 |
| 25* | siltstone concretions (S) | 3 | 7 | 12 | 170 | 76 | 0.08 | 341 | 54 | 80 | 5 |
| 24* | sandstone concretions (S) | 3 | 8 | 15 | 170 | 76 | 0.30 | 343 | 50 | 193 | 3 |
| 23 | sandstone concretions (S) | 3 | 8 | 10 | 170 | 76 | 0.82 | 330 | -21 | 18 | 12 |
| 22 | sandstone concretions (S) | 3 | 7 | 12 | 170 | 76 | 1.43 | 336 | -34 | 31 | 8 |
| 21* | porphyritic diorite sill | 3 | 9 | 15 | 170 | 76 | 0.58 | 156 | -50 | 15 | 10 |
| 20 | sandstone concretions (S) | 3 | 5 | 9 | 170 | 76 | 0.54 | 339 | -11 | 60 | 7 |
| 19 | sandstone concretions (S) | 3 | 6 | 8 | 170 | 76 | 2.35 | 352 | -16 | 11 | 18† |
| 18* | diorite sill | 3 | 7 | 12 | 170 | 90 | 1.01 | 149 | -39 | 32 | 8 |
| 17* | mg sandstone (S) | 4 | 7 | 8 | 173 | 89 | 3.07 | 341 | 54 | 18 | 13 |
| R4 | sandstone (S) | 3 | 7 | 13 | 173 | 89 | 1.04 | 2 | -34 | 7 | 17† |
| 16 | fg sandstone (S) | 5 | 9 | 16 | 173 | 89 | 3.36 | 346 | -11 | 20 | 8 |

2156220b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | siltstone (S) | 5 | 8 | 14 | 173 | 89 | 4.34 | 328 | -22 | 33 | 7 |
| | *Tatlow Syncline, Section 2 of North Limb (51.4°N, 123.9°W)* | | | | | | | | | | |
| 54 | porphyritic diorite dike | | 5 | 7 | 140 | 72 | 12.97 | 034 | 49 | 16 | 16† |
| 53 | andesite flow (P) | 1 | 7 | 9 | 140 | 72 | 62.81 | 297 | -19 | 75 | 6 |
| 52 | andesite flow (P) | | 5 | 8 | 140 | 72 | 73.60 | 297 | -16 | 141 | 5 |
| 51 | andesite flow (P) | | 5 | 7 | 140 | 72 | 67.64 | 306 | -16 | 48 | 9 |
| 50 | andesite flow (P) | | 6 | 9 | 140 | 72 | 68.01 | 297 | -01 | 89 | 6 |
| | *Tatlow syncline, isolated sites of north limb (site 60, 51.3°N, 123.9°W)(site 62, 51.4°N, 123.9°W)* | | | | | | | | | | |
| 60‡ | basalt (P) | 3 | 6 | 10 | 187 | 14 | 0.45 | 015 | 71 | 85 | 5 |
| 62 | sandstone (S) | 3 | 5 | 8 | 180 | 74 | 6.37 | 344 | 8 | 17 | 14 |
| | *Davidson Bridge (51.6°N, 123.7°W)* | | | | | | | | | | |
| 29 | sandstone concretions | 3 | 5 | 7 | 323 | 27 | 25.03 | 311 | 72 | 20 | 14 |

Here *h*, *c*, and *s* are number of hand samples, cores, and specimens, respectively. *BDDA* and *BD* are bedding downdip azimuth and bedding dip of site sampled or it's nearest neighbor; *M* is the mean intensity of remanent magnetization; $D_H$, $I_H$ are mean site directions (declination, inclination) relative to present horizontal; *k* is a precision parameter [*Fisher*, 1953]; $\alpha_{95}$ is radius of 95% confidence circle. For rock types: fg, fine grained; vfg, very fine grained; mg, medium grained; pl lam, plane laminated; cross lam, ripples or cross bedding. S, Silverquick conglomerate; P, Powell Creek volcanics. Within sections sites are listed in inverse stratigraphic order: south limb site 1 is lowest, site 14 the highest; north limb, section 1, site 28 is lowest site 15 the highest; north limb, section 2 site 54 is lowest, site 50 the highest. R4 and R5 are reconnaissance sites collected by J. A. Maxson and K. L. Kleinspehn in 1988.

* Sites with syndeformational magnetizations.

† Sites with $\alpha_{95} > 15°$ omitted from summary calculation.

‡ Present Earth's field only recorded.



21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

attitudes were determined from interbedded, planar-laminated fluvial strata exposed within a few meters of each sampled flow. These interbeds provide accurate estimates of paleohorizontal for the lavas.

The Silverquick conglomerate contains rare Albian to Cenomanian plant fossils. An older age limit is provided by late Albian marine fossils in the underlying Taylor Creek Group [*Jeletzky and Tipper*, 1968; *Tipper*, 1969]. Fission track analyses of Silverquick detrital zircons, collected immediately southeast of the Tatlow syncline, indicate a post-late Albian age [*Garver*, 1989, 1992; *Garver and Brandon*, 1994]. A younger limit to the age of the Silverquick conglomerate is provided by an $^{40}Ar/^{39}Ar$ date of 92 ± 1.3 Ma (Cenomanian, using the timescales of *Kent and Gradstein* [1986] and *Cande and Kent* [1992]) from the base of the overlying Powell Creek volcanics on the north limb of the Tatlow syncline (Maxson et al., unpublished data) Thus we regard the Silverquick conglomerate as Cenomanian in age. The youngest Powell Creek volcanics, sampled near the syncline axis, gave an $^{40}Ar/^{39}Ar$ date of 79 ± 4.1 Ma (Campanian) (Maxson et al., unpublished data). Thus all sedimentary and volcanic samples collected from the Tatlow syncline have depositional ages of Cenomanian through Campanian.

Small hypabyssal intrusions cut the mid-Cretaceous sequence. Four of them have been sampled, and one dike (site 54, Figure 2) yielded a Maastrichtian $^{40}Ar/^{39}Ar$ age of 69.9 ± 3.2 Ma (Maxson et al., unpublished data). The intrusions may not all be coeval.

We sampled both limbs of the Tatlow syncline, between Chilko Lake and the Taseko Lakes and the Taseko and Tchaikazan faults (Figure 2). The syncline, which is about 20 km wide, is a coherent structure, although it is interrupted by faults with small displacements [*McLaren*, 1987; *Riddell et al.*, 1993a, b; *Umhoefer and Kleinspehn*, 1995] which may have produced small block rotations. Plunge is $<4°$ to the east. Initial folding may be related to the Late Cretaceous Eastern Waddington thrust system [*Rusmore and Woodsworth*, 1991; *Umhoefer and Kleinspehn*, 1995]. Site 29 lies outside the Tatlow syncline, north of the Yalakom Fault (Figure 2) in late Cretaceous strata that have been correlated to the Silverquick conglomerate [*Hickson et al.*, 1991], but are likely coeval with the youngest Powell Creek volcanics (Maxson et al., unpublished data).

## Procedures

Six to nine cores or two or more oriented hand samples were taken from a single flow, intrusion or sedimentary bed, referred to as a "site" (Table 1). Orientation was by magnetic compass and by solar compass when the Sun shone (75% of samples). The two methods generally agreed to within 2°. Usually, three cores were drilled in the laboratory from each hand sample. Two specimens were cut from most cores. The natural remanent magnetization (NRM) of specimens was measured using automated Schonstedt magnetometers. In pilot studies, we demagnetized two specimens from each site in 10 to 15 steps, one thermally using a Schonstedt TSD1 furnace, and one by alternating magnetic field (AF) using either a Sapphire SI4 or a Schonstedt GSD5 demagnetizer. To monitor possible chemical changes caused by heating, we measured magnetic susceptibility during thermal runs using a Sapphire SI2 instrument. From these pilot studies, we determined the demagnetization steps required to define the magnetizations present in the remaining specimens.

Principal component analysis (PCA) was used, typically by line fitting through three or more points. In any one specimen, no more than two magnetizations were present: a soft magnetization aligned approximately along the present Earth's field (PEF) considered to be a viscous remanent magnetization. (VRM) acquired since the last reversal (Brunhes overprint), together with a magnetization of higher stability. At five sites the final stages of demagnetization could not be observed and hence PCA could not be used, either because the specimens underwent chemical changes producing new magnetic minerals at temperatures as low as 500°C (sites 16, 22, and 23) or because the magnetization changed little in intensity or direction during increasing AF demagnetization (sites 26 and 27). In addition, 11 specimens (out of 58) from five sites (4, 8, 15, 51, and 53) exploded during thermal demagnetization (≈ 400°C), presumably because of zeolites, and their magnetizations were determined from single demagnetization steps, typically between 300° and 500°C; this procedure is justified because Brunhes overprints are effectively removed at these temperatures in all other specimens.

Out of 437 specimens studied, 42 were rejected for the following reasons: 11 had inhomogeneous magnetization with low internal precision $k_s < 5$, calculated from four determinations of direction made from 12 readings of the $X$, $Y$ and $Z$ components [e.g., *Harrison*, 1980]; 13 had line-fitting errors >20°; and 18 had directions greater than two standard deviations from the mean site direction.

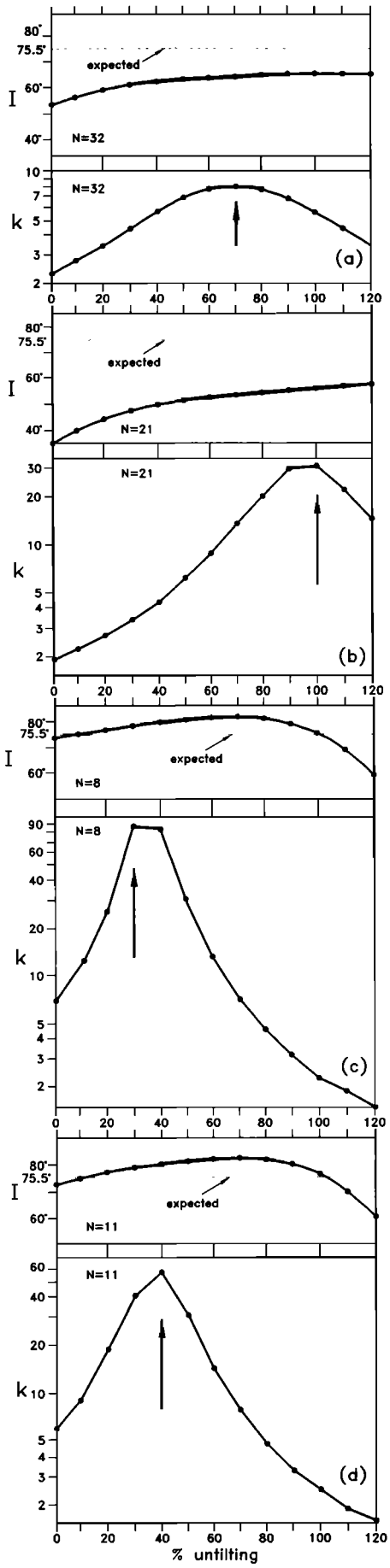
Mean site directions are listed in Table 1. Except for site 60, all site mean directions departed significantly from the present Earth's field. Within-site directions were usually well grouped, and uncertainties ($\alpha_{95}$) in site mean directions were typically <10°. However, six sites (asterisks in Table 1) had $\alpha_{95} > 15°$ and were considered too inaccurate to be included in the final analysis.

## Recognition of Pretilting and Syntilting Magnetizations

We applied *Fisher*'s [1953] statistics and calculated the inclination of mean direction and precision as a function of degree of untilting. For all 32 sites the inclination is always systematically less than expected, and the precision is maximum after 70% untilting (Figure 3a). In Figure 3b, 21 sites with high $T_{UB}$ (described later) have systematically lower inclinations with maximum precision at 100% untilting; these magnetizations were acquired before tilting. In Figure 3c eight sites from the two units with low $T_{UB}$ have inclinations close to that expected and maximum precision at 30% untilting; these magnetizations were acquired during tilting. Adding, to the latter subset, data from three diorite intrusions yields essentially the same result (Figure 3d).

Clearly, two magnetizations are present. The more common has directions that are widely scattered when uncorrected for tilt and become best grouped after correction for the full amount of tilt. The other magnetization has directions that are best grouped after partial correction for tilting. The former are described first.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License




## Pretilting Magnetization

### Silverquick Sedimentary Strata

Thermal demagnetization generally gave distributed $T_{UB}$ ranging up to 575°C. Below 300°C, directions of magnetizations moved a few degrees away from the PEF during removal of small overprints acquired since the last reversal (Bruhnes overprints) (Figure 4). Between 300° and 500°C, directions formed a cluster or endpoint, and the removed magnetization (RM) obtained by line fitting in this range had a direction (uncorrected for bedding tilt) to the southeast with moderate downward inclinations. Above 550°C, directions scattered and intensities increased. During heating, the susceptibility changed little up to about 550°C and thereafter increased dramatically, signifying that new magnetic minerals were being made, presumably from the breakdown of mafic minerals; this behavior occurred in most sedimentary specimens (Figure 5).

Results from planar-laminated siltstones and concretions are compared in Figures 6-9. Directions (uncorrected for tilt) pointed to the southwest with intermediate downward inclination after removal of small Bruhnes overprints. Thereafter, intensities decayed linearly to the origin. Magnetizations removed along the straight-line segment of decay plots have similar directions for both rock types for both AF and thermal treatment.

Study of polished thin sections showed that magnetite is by far the most common opaque mineral present. Its grain size varies widely, and it is this variability that we presume is responsible for the distributed $T_{UB}$ [*Dunlop*, 1986] which approach the Curie temperature of magnetite (200° to 575°C, Figures 4, 7, and 9) and the wide range of coercivities of remanence (<10 to 100 mT, Figures 6 and 8). Magnetite occurs as homogenous subhedral to anhedral grains range in size from 30 μm down to 3 μm and probably below (we are unable to identify optically grains < 3 μm); these we consider detrital. Magnetite also is present as very fine-grained pore lining, as thin coatings on silicate sand grains, and as alteration products along the margins and cleavage fractures of mafic minerals; these we consider authigenic. Finally, magnetite occurs as small (3 to 5 μm) euhedra within 1 to 3 mm lithic fragments. Minor amounts of pyrite and rare traces of hematite (lining pores and associated with magnetite) probably are also authigenic. Possible processes of magnetization are therefore (1) detrital remanent magnetization (DRM) or postdepositional DRM (grain alignment in the field after deposition but prior to cementation) carried by detrital magnetite; and (2) chemical remanent magnetization (CRM) carried by authigenic magnetite deposited during cementation. It is unlikely that magnetite in lithic fragments played a role because the

**Figure 3.** (opposite) Inclination of mean directions and precision calculated by Fisher statistics versus degree of untilting. (a) All sites. (b) Sites with high $T_{UB}$. (c) Sites with low $T_{UB}$. (d) Sites from diorite intrusions added to Figure 3c. Untilting of 0% corresponds to beds in their present positions and 100% corresponds to beds fully restored to paleohorizontal using panel tilts given in footnotes to Table 2. Arrows denote maximum precision.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

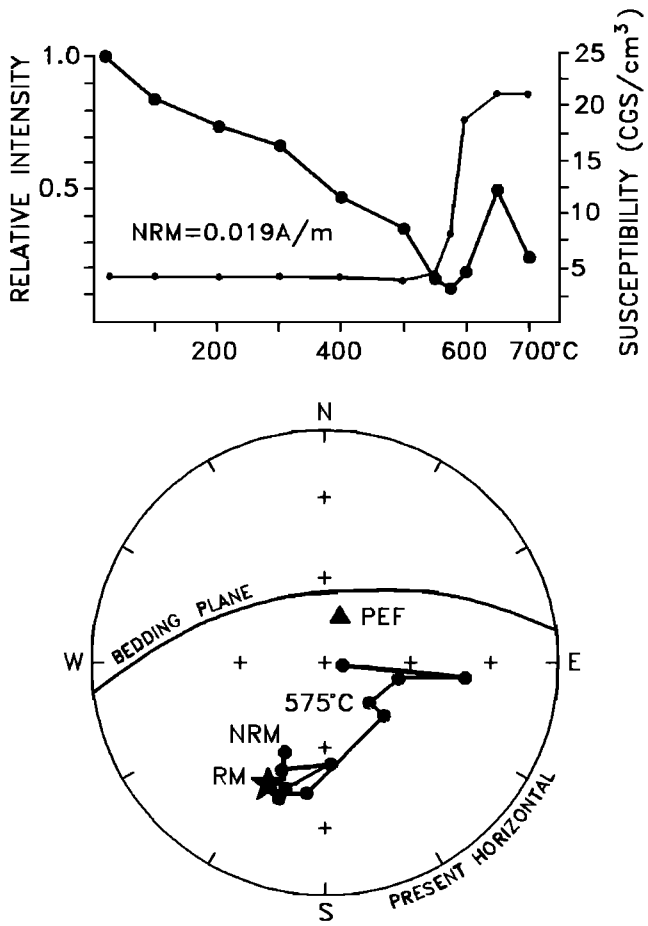



**Figure 4.** Thermal demagnetization of remanent magnetization and magnetic susceptibility of concretion in lithic sandstone, site 7, Silverquick conglomerate. (top) Changes in intensity (large dots) and susceptibility (small dots). (bottom) Changes in direction, plotted with respect to present horizontal on an equal-area projection, solid (open) symbols denoting downward (upward) directions. NRM, direction of the natural remanent magnetization. RM (star), direction of magnetization removed between 300° and 500°C (calculated by principal component analysis) PEF direction of Earth's present geomagnetic field. Solid line shows the trace of the bedding plane on lower hemisphere.

fragments probably were too large to be aligned magnetically during deposition.

## Powell Creek Lava Beds

AF demagnetization yielded excellent end points and orthogonal plots linear to the origin (Figure 10). Coercivities commonly exceeded 100 mT. Thermal demagnetization yielded $T_{UB}$ in the range 500° to 575°C, typical for fine-grained titanium-poor magnetite. Sometimes a small component with the same direction remained above 575°C, carried presumably by hematite (Figure 11). Removed magnetizations, obtained by line fitting in the range 500° to 650°C or 20 to 60 mT, were in excellent agreement and directed toward the northwest with shallow negative inclinations. Susceptibility did not change during heating (Figure 5).

In polished thin section, abundant subhedral to euhedral magnetite grains (typically about 25 μm) contain delicate ilmenite exsolution lamellae and subgrains commonly in a mottled groundmass of finely intergrown magnetite and ilmenite (class III or IV of *Wilson et al.* [1968]; see Figure 12). The textures are likely produced by cooling from high temperatures under oxidizing conditions. We suggest that the stable magnetization is a thermoremanent magnetization (TRM), carried by magnetite within these finely divided composite grains. The sampled lava beds are fresh, plagioclase is only weakly altered, and mafic minerals are not chloritized. Although pyrite or hematite was not identified, trace amounts of the latter must be present to carry the small magnetization observed with $T_{UB} > 575°$ (Figure 11). A sill (site 10) has magnetizations like those in the stratified beds and therefore is considered co-magmatic with the Powell Creek volcanics.

We sampled a fifth site (site 60) close to the axis of the syncline (Figure 2) to assess any latitudinal changes that may have occurred during the approximately 12 m.y. history of Powell Creek volcanism, a time range based on $^{40}Ar/^{39}Ar$ ages (Maxson et al., unpublished data). Unfortunately, the rock has low coercivities (only 10% of intensity remains at 40 mT), low $T_{UB}$ (100 to 400°C), and their mean direction coincided with the PEF (Table 1). We considered the magnetization at site 60 to be a Bruhnes overprint.

## Untilting Test

Mean site directions are listed in Table 1. Uncorrected for bedding tilt, they are widely scattered, but after correction the grouping is good, and directions point northwest with intermediate downward inclinations (Figure 13). Polarity is normal, as expected for rocks magnetized during the Cretaceous normal superchron (120 to 83 Ma). Mean inclinations of both sedimentary ($56.8 \pm 4.8°$, $N = 16$ sites) and igneous ($51.5 \pm 8.0°$, $N = 5$ sites) rocks are in good agreement. It is noteworthy that after correction for

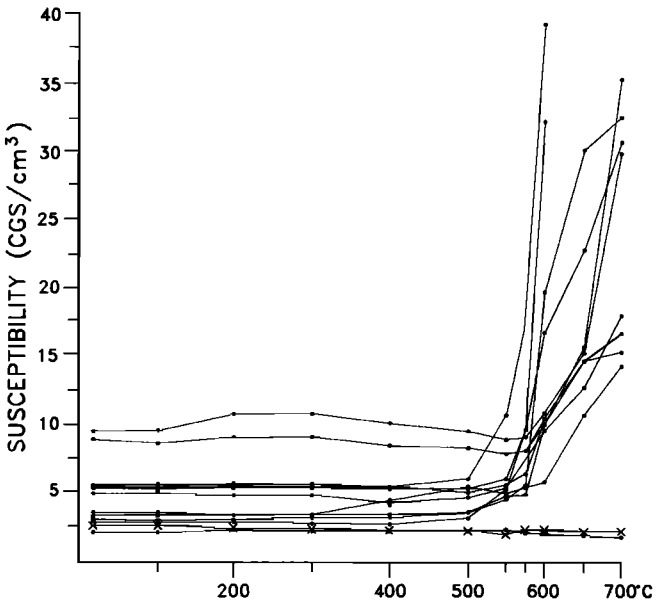



**Figure 5.** Changes in magnetic susceptibility during thermal demagnetization of 10 specimens of Silverquick sedimentary rocks (dots) showing increases above 550° or 575°C. Specimens from site 51 (crosses) Powell Creek volcanics and site 29 (dots) Silverquick concretion show no increase.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

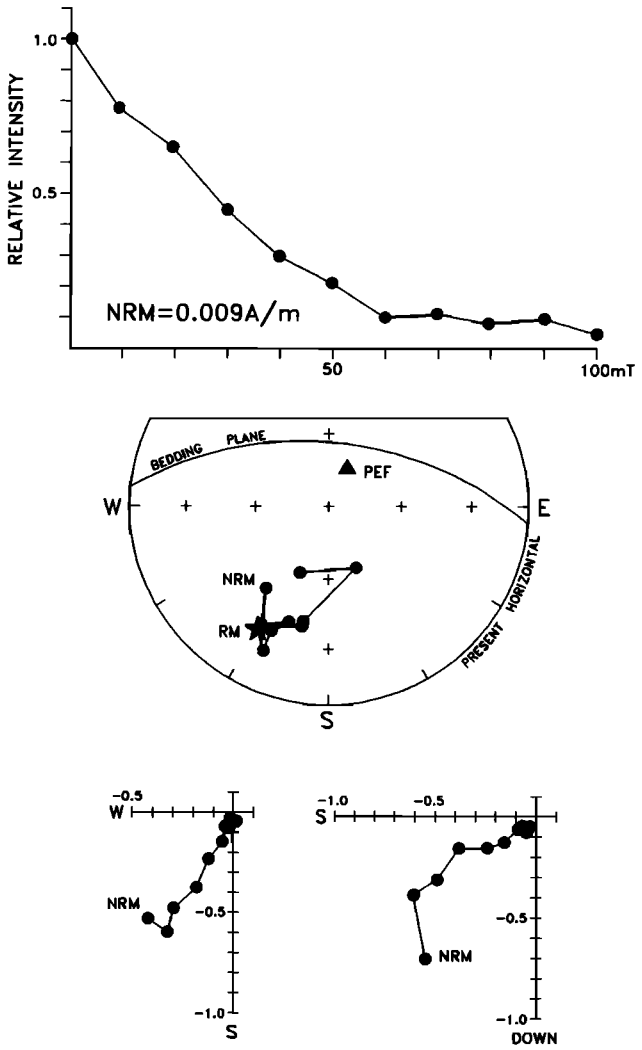



**Figure 6.** AF demagnetization of plane-laminated siltstone, Silverquick conglomerate, site 8. (top) Changes in intensity. (middle) Changes in directions. RM (star), direction of magnetization removed between 10 and 60 mT. (bottom) Normalized orthogonal plots. Symbols and conventions as in Figure 4.

bedding tilt, site 29, situated northeast of the Yalakom Fault (Figure 2), has a direction which falls within the group of directions observed from the Tatlow syncline (Figure 13 and Table 2).

Mean directions of magnetization of the three main sections (Table 2), uncorrected for bedding tilt, are 128° apart (Figure 14). After tilt correction, the means of section 1 of the north limb and the south limb coincide. The mean inclination of section 2 of the north limb agrees with that of the other two localities, but its declination differs by 39°. The strikes of beds in the two sections also differ by a comparable amount (Table 2), indicating that the small structural block containing section 2 has rotated 30 to 40° counterclockwise about a vertical axis, relative to the main sections.

Because of the possibility of rotations among localities, conventional Fisher's statistics cannot, strictly speaking, be applied to the data as a whole. However, because only four sites are involved, the error incurred by doing so will be small. To obtain better estimates, we applied blockrotation Fisher analysis (BRF [*Monger and Irving*, 1980]) which involved rotating all mean site directions, corrected for bedding, about a vertical axis so that the mean declination of each section (rows 1, 3, and 5 in Table 2) is zero. A Fisher average of the rotated sites is then calculated. BRF analysis was applied first to data (19 sites) from sections on the north and south limbs of the Tatlow syncline and a mean inclination of 55.4° ± 3.4° ($k$ = 55) was obtained. Adding data from two isolated sites (29 and 62 for a total of 21 sites) yielded, after correction for bedding tilt, a mean inclination of 55.4° ± 3.4° ($k$ = 48).

Figure 15 shows the effect of incremental untilting and clearly demonstrates that magnetizations were acquired prior to folding. Mean inclination ($I_m$) increased as untilting proceeded, becoming constant at about 55° between 80 and 120% untilting. Dispersion, expressed as the circular standard deviation $\Theta_{63}$, reaches a minimum at 100% untilting, and the precision ($k$) ranges from 5 before untilting

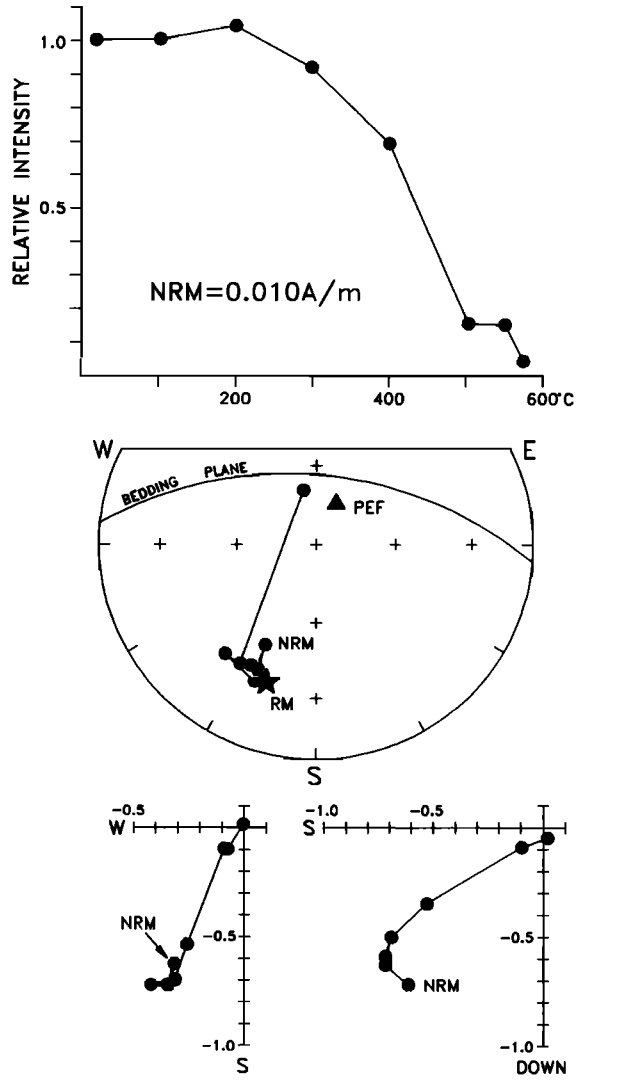



**Figure 7.** Thermal demagnetization of plane-laminated siltstone, Silverquick conglomerate, site 8. RM (star), direction of magnetization removed between 300° and 500°C. Symbols and conventions as in Figure 4.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

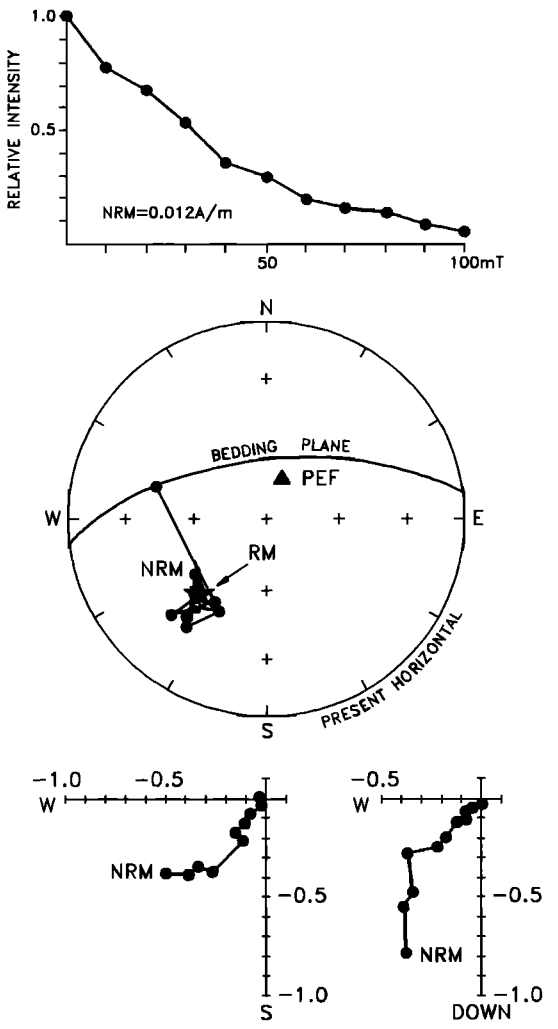



**Figure 8.** AF demagnetization of lithic sandstone concretion, Silverquick conglomerate, site 7. (top) Changes in intensity. (middle) Changes in directions. RM (star), direction of magnetization removed between 10 and 100 mT. (bottom) Normalized orthogonal plots. Symbols and conventions as in Figure 4.

to a maximum of 48 at 100% untilting, a slight improvement from Fisher analysis (Figure 3b).

For comparison we then used the inclination-only, maximum likelihood analysis of *McFadden and Reid* [1982], which yielded the same trends with incremental untilting as BRF. Inclination increased as untilting proceeded, reaching a plateau at 56° between 50 and 130% untilting, and maximum precision was achieved at 100% untilting ($k$ = 34). For 21 sites, $I_m$ is 56.3 ± 5.4° which is not significantly different from that obtained by BRF analysis but the error is greater as expected because declination data are not used.

### Age and Origin

The rocks are Cenomanian to Campanian in age. Their magnetic polarities are all normal, which indicates that the magnetization was probably acquired before the end of the Cretaceous normal superchron (83 Ma [*Cande and Kent*, 1992]), not later in the Cretaceous when the paleofield reversed frequently. Because the lava beds preserve original crystallization textures (Figure 12) and have a prefolding magnetization of high stability (Figures 10 and 11), we interpret their magnetizations to be TRM acquired during late Cenomanian cooling. Because of the magnetic stability (Figures 6 to 9) and the petrographic characteristics of the sedimentary rocks, we interpret their pretilting magnetization to be a combination of DRM or postdepositional DRM, and CRM acquired during early cementation, soon after deposition in Cenomanian time. Hence pretilting magnetizations probably were acquired in the interval 94 to 83 Ma (Cenomanian to earliest Campanian).

## Origin of Anomalously Shallow Inclinations of Pretilting Magnetization

The close agreement, already noted, between the mean inclination of igneous and sedimentary sites is strong evidence that prefolding magnetizations accurately record the paleofield. However, the mean inclination is 20° shallower than the 75.5° ± 2.1°, expected from observations from mid-Cretaceous cratonic rocks (see below). Processes that could have shallowed the inclinations are now examined.

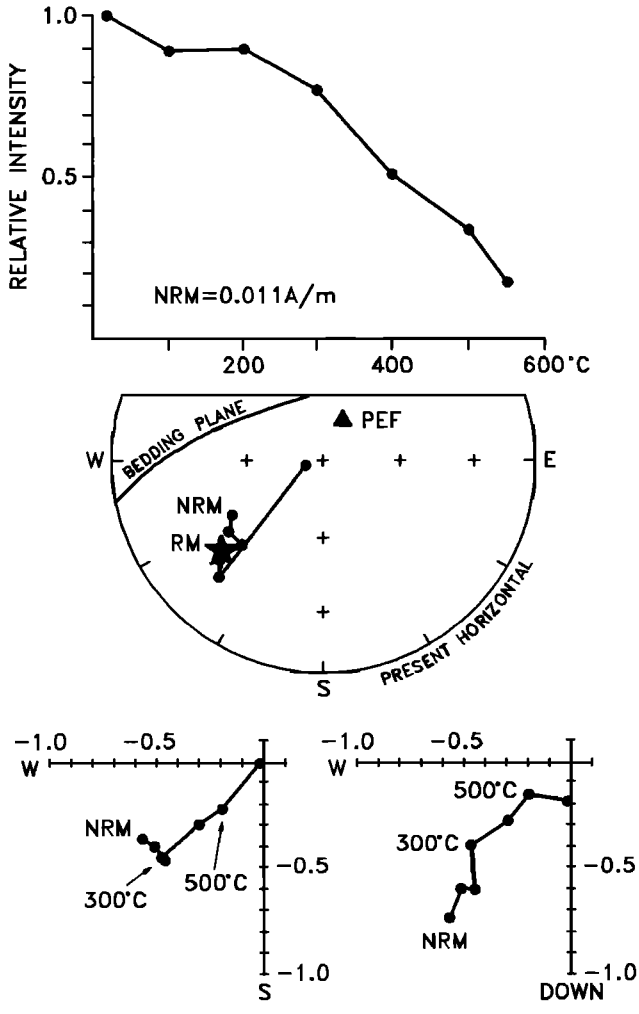



**Figure 9.** Thermal demagnetization of lithic sandstone concretion, Silverquick conglomerate, site 7. RM (star), direction of magnetization removed between 300° and 500°C. Symbols and conventions as in Figure 4.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

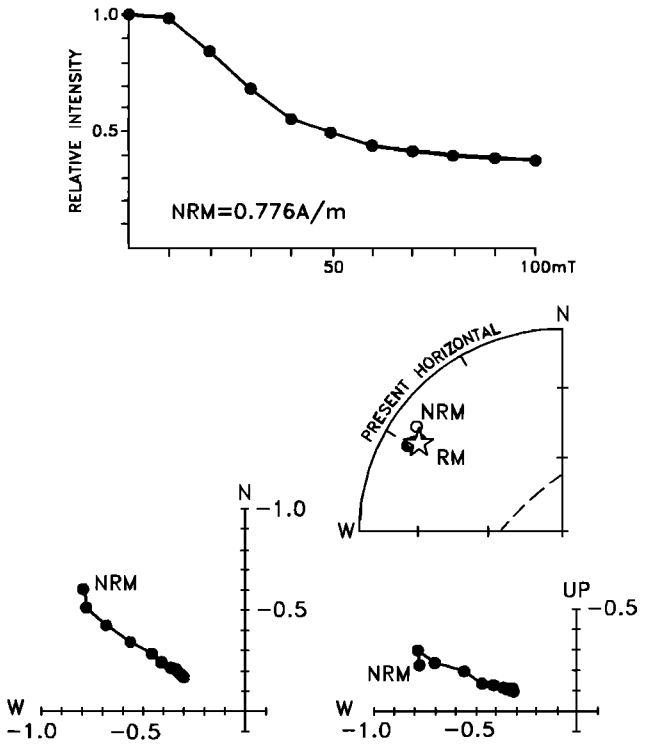



**Figure 10.** AF demagnetization of andesite of the Powell Creek volcanics, site 52. RM (star), direction of magnetization removed between 20 and 60 mT. Dashed line shows the trace of the bedding plane (dips down to the SE) on the upper hemisphere. Symbols and conventions as in Figure 4.

If magnetization is carried by elongate grains magnetized along their long axis, then compaction may have rotated the grains toward bedding planes and shallowed inclination. Magnetization of thinly bedded overbank silt and fine-grained sand would be especially prone to such effects. In fine-grained strata of the Silverquick conglomerate, as previously noted, bedding is draped around concretions, indicating that they were lithified before the surrounding sediments and were therefore largely protected from the effects of compaction. In spite of this, concretions actually record a shallower mean inclination (52.8° ± 6.1°, $N = 8$ sites) than the ambient sandstone and siltstone beds (59.8° ± 9.6°, $N = 8$ sites), although the difference is not significant. Lava bed samples also have a mean inclination (52.8° ± 10.8°, $N = 4$) essentially identical to that of concretions. Moreover, inclinations of sediments of differing grain size are not significantly different (sandstone 56.7° ± 7.0°, $N = 12$ sites; siltstone 55.0° ± 10.8, $N = 4$ sites), and their demagnetization characteristics are similar for both strata containing and lacking bedforms (Figures 6 to 9). Systematic alignment of elongate iron minerals was not observed in polished thin section. Furthermore, if, as the demagnetization and petrographic studies show, the magnetization is carried by disseminated, equant, presumably detrital magnetite, and/or by authigenic magnetite, then such grains, because of their shape, are unlikely to have been rotated systematically during compaction. For these reasons we reject the hypothesis that compaction has significantly shallowed inclination.

Lava beds might bear a shallow inclination because of alignment of elongate iron minerals during flow. If alignment was present, the rocks should show appreciable magnetic anisotropy. The anisotropy of susceptibility in four lava specimens was very small (the ratio of maximum to minimum susceptibility ranges from 1.03 to 1.06). A field of 180 mT was applied along the cylindrical axes of the same specimens to produce an isothermal remanent magnetization (IRM). This was then incrementally demagnetized. The inclinations ranged from 90° to 87°, not significantly different from the expected 90°, and remained unchanged during demagnetization up to 30 mT; there is no significant anisotropy of IRM.

The observed shallow inclination could have been produced if the rocks had been deposited at their present latitude relative to cratonic North America but on a paleoslope dipping 20° to the north-northwest. Restoration of bedding to paleohorizontal then would result in overrotation and spurious shallow inclinations. This hypothesis is rejected for four reasons. Paleocurrent indicators suggest that the paleoslope dipped not to the north-northwest but to the south-southeast. Slopes of modern gravelly meandering fluvial systems with cohesive banks are only 1° to 2° [*Church and Rood*, 1983]. Although intermediate lavas are often deposited on slopes of 15° to 33° [*Cas and Wright*, 1987], we reiterate that the bedding orientations for the Powell Creek volcanics were not measured from flows but from planar-laminated fluvial deposits intercalated between them. The shallow magnetic inclinations would require a paleoslope that was uniform over approximately 25 km (the

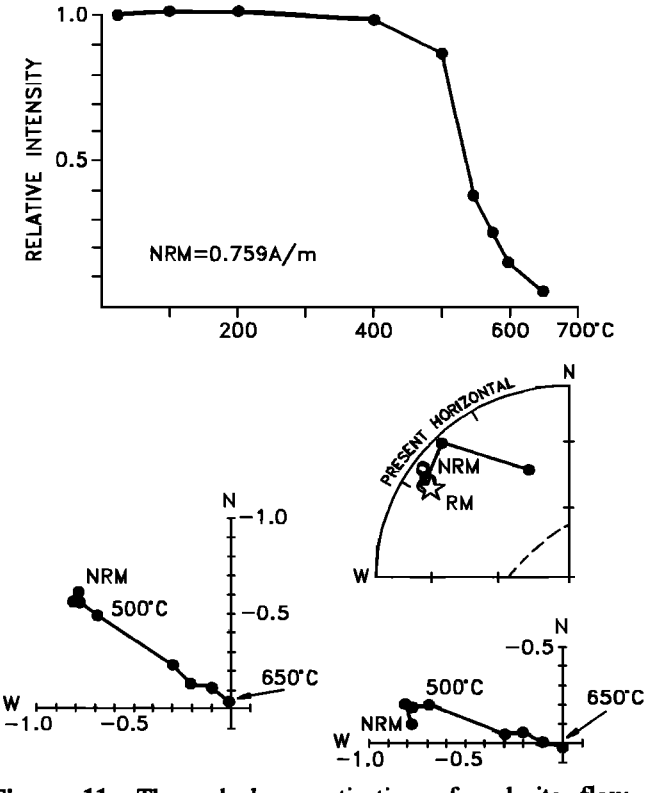



**Figure 11.** Thermal demagnetization of andesite flow, Powell Creek volcanics, site 52. RM (star), direction of magnetization removed between 500° and 650°C. Symbols and conventions as in Figure 4.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

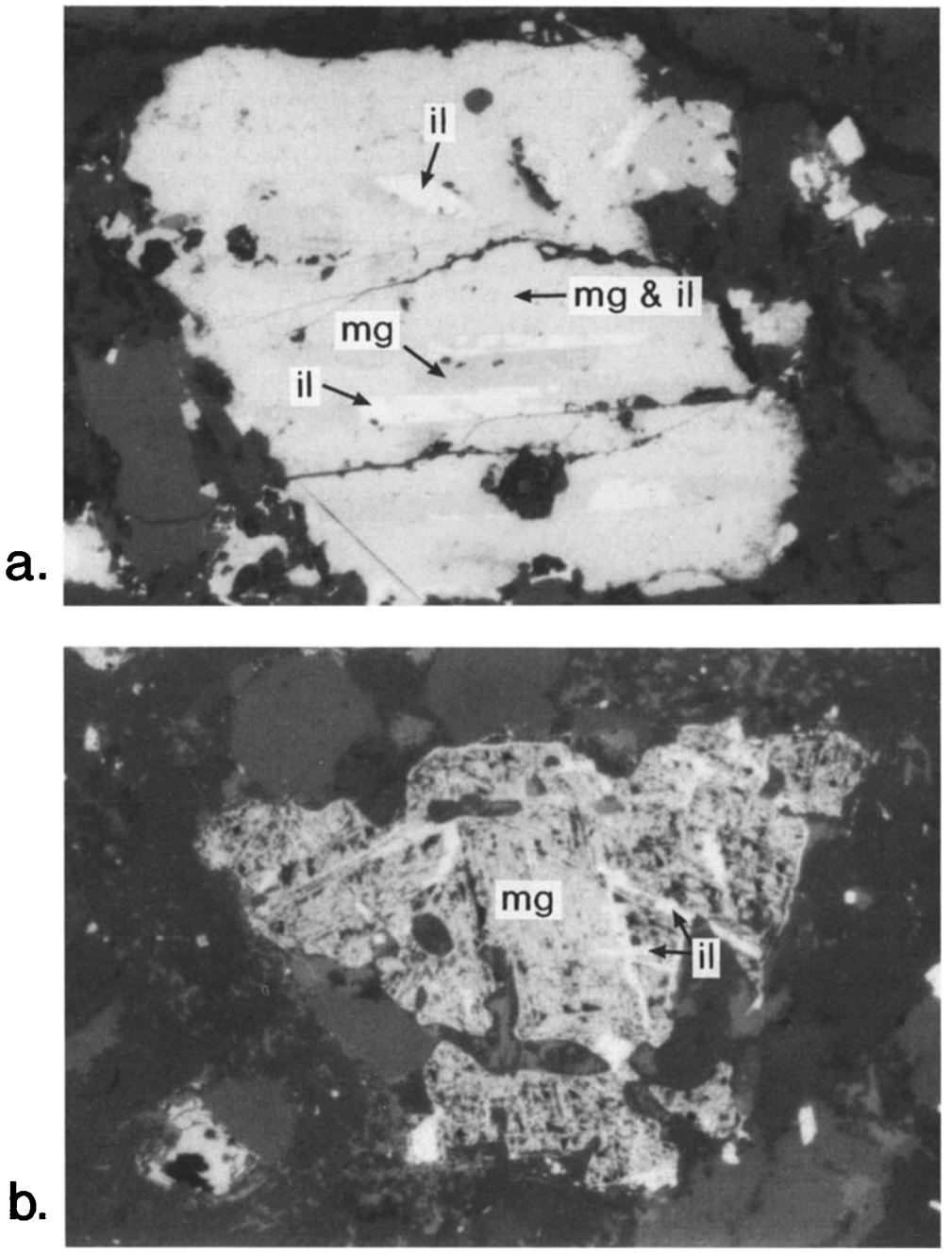



**Figure 12.** Photomicrographs of polished thin sections taken in polarized reflected light, fields of view are 20 μm. (a) Andesite, site 51, subhedral crystal with ilmenite exsolution lamellae (highest reflectivity) and patches and lamellae of magnetite (lowest reflectivity) in a groundmass of very finely intergrown magnetite and ilmenite (medium reflectivity and mottled or mossy texture). (b) Andesite, site 50, subhedral magnetite crystal with ilmenite exsolution lamella (high reflectivity). Black triangular areas are holes plucked out during polishing.

restored geographic spread of sampling sites, assuming a cylindrical fold shape (Figure 2b) and that remained consistent during deposition of 1200 m of strata. Such uniformity seems improbable.

An inclination error could have been produced if the rocks were deposited essentially where they are today relative to cratonic North America, uniformly tilted down 20° to the north-northwest, then magnetized, and subsequently folded. In this hypothesis, tilting would have occurred prior to acquisition of magnetization which exclude a DRM, for which, as we have argued above, there is substantial evidence. Magnetizations are limited by this hypothesis to those formed by low temperature chemical processes (chemical remanent magnetization, CRM), such as low-temperature hydrothermal alteration of magnetite, or precipitation of magnetite and hematite from groundwater or igneous fluids [e.g., *Hall and Fisher*, 1987]. We can rule out a posttilting TRM because the required heating to 400 or

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

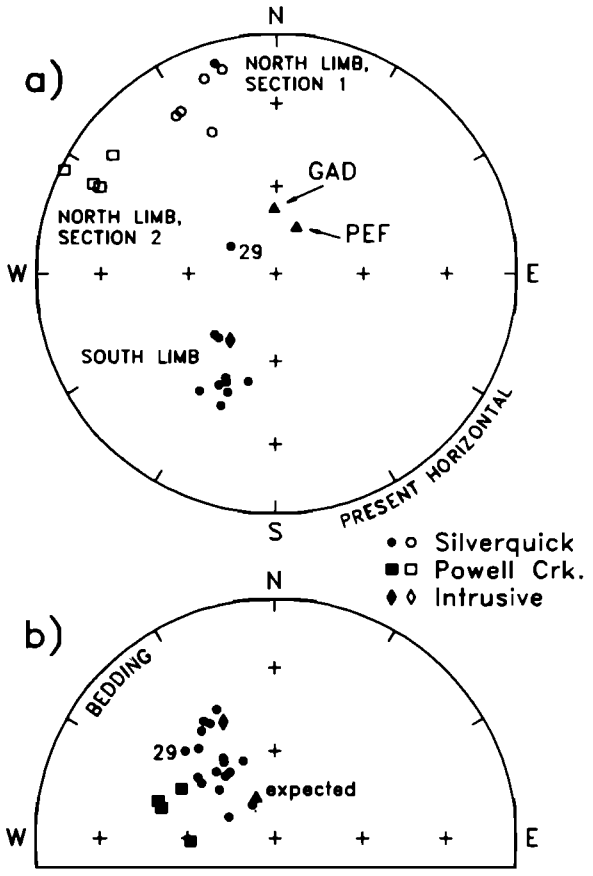



**Figure 13.** Mean site directions of prefolding magnetization: (a) uncorrected for tilt and (b) after 100% bedding correction (see footnotes to Table 2). Sites with $\alpha_{95} > 15°$ not plotted. Expected, mid-Cretaceous mean field direction expected [*Van Fossen and Kent*, 1992]; GAD, geocentric axial dipole field direction; PEF, present Earth's field direction. Open (solid) symbols are upward (downward) directions.

500°C or more (illustrated by $T_{UB}$, Figures 7, 9, and 11) would have metamorphosed the rocks to high greenschist or amphibolite facies, very much higher than the observed zeolite facies. If a posttilting CRM were solely responsible for the magnetization, it must have affected both sedimentary and igneous rocks in a remarkably uniform way. But the sedimentary and igneous rocks have very different porosities and permeabilities, and there is no reason to expect similar diagenetic histories. Moreover, oxide grains in the flows have delicate ilmenite and magnetite exsolution lamellae and intergrowths (Figure 12), textures that would not have been preserved intact had there been a pervasive CRM, as *Hall and Fisher* [1987] have shown. There is no petrological evidence of alteration in the flows other than the deuteric, high-temperature alteration to be expected upon cooling following extrusion [*Wilson et al.*, 1968].

Finally, some might argue that the shallow inclination is the product of a perturbation of the geomagnetic field, tantamount to saying that the average paleofield is not predominantly dipolar. Paleopoles from cratonic North America are exceedingly well grouped (circular standard deviation is 4°, see below), showing that the average paleofield on the craton conforms very well to that of a geocentric dipole [e.g., *Van Fossen and Kent*, 1992]. Cretaceous rocks from the western elements of the Cordillera, extending over a distance of 3500 km from Baja California to Alaska, have yielded paleopoles that without exception, diverge typically by >20° from cratonic paleopoles [*Beck*, 1976, 1980, 1991, 1992; *Irving*, 1979; *Irving and Wynne*, 1990]. If this phenomenon had its origin in the geomagnetic field, then, because of its scale, its source must have been at least as deep as the outer core. Physically it is inconceivable that such a source did not also affect the paleofield of the nearby craton. It could be argued that the perturbation of the Cordilleran paleofield was short-lived and hence that its chance of being observed from Cretaceous rocks on the craton is slim. This is not so, because aberrant magnetizations have been observed from the western elements of the Cordillera in rocks ranging in age over 30 m.y. [e.g., *Irving and Wynne*, 1990]. Thus we reject the suggestion that perturbations of the paleofield are responsible for the observed shallow inclinations.

Because we can rule out compaction, grain alignment during lava flow, high initial paleoslope, tilting prior to magnetization and gross perturbations of the paleofield as explanations of the observed shallow inclinations, we conclude that the pre-folding magnetizations observed in the Silverquick and Powell Creek strata provide an accurate record of the inclination of the paleofield and hence of latitude in early Late Cretaceous time.

## Younger Magnetizations

As already noted, 11 sites have syntilting magnetizations (Figure 3). Seven sites are from the Silverquick sedimentary strata, one is from Powell Creek basaltic breccia and three are from hypabyssal diorite intrusions (Table 1). A fourth intrusion (diorite dike, site 54) has been dated by the $^{40}Ar/^{39}Ar$ method as 69.9 ± 3.2 Ma or Maastrichtian (Maxson et al., unpublished data) but did not give acceptable paleomagnetic data ($\alpha_{95} > 15°$).

Bedded rocks with syntilting magnetizations are highly altered. Iron-magnesium silicates and feldspars are partially or totally replaced by calcite. The directions carried by these sites are all normal. The magnetizations have very high coercivities (>100 mT), low $T_{UB}$ (200° to 400°C), linear decay to the origin, and excellent end-points (Figures 16 and 17), characteristics typical of pyrrhotite, in contrast to the prefolding magnetization which is carried by magnetite. Pyrrhotite, however, was not observed in polished thin sections, but pyrite was present as 10 to 20 μm framboidal knots or polycrystalline aggregates. Pyrrhotite presumably occurs as ultrafine particles intimately associated with the pyrite. In similar occurrences elsewhere, the two minerals are interpreted as coexisting diagenetic phases with pyrrhotite formed as a stage in the pyritization process [*Roberts and Turner*, 1993].

Rocks from the hypabyssal intrusions also are altered. Amphiboles are partially (sites 13 and 21) or totally (site 18) replaced by calcite and chlorite, and plagioclase is weakly to strongly saussuritized. Magnetite occurs as fine (<3 μm) disseminations, 20 μm homogeneous euhedra, and, at site 13, 200-300 μm subhedral grains with exsolution lamellae and irregular domains of ilmenite (class III oxidation state of

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 2.** Results by Localities

| | $N$ | $D_h$, deg | $I_h$, deg | $k$ | $\alpha_{95}$, deg | $\alpha_{63}$, deg | $D_b$, deg | $I_b$, deg | $k$ | $\alpha_{95}$, deg | $\alpha_{63}$, deg |
|---|---|---|---|---|---|---|---|---|---|---|---|
| South Limb | | | | | | | | | | | |
| 1. Silverquick conglomerate | 10 | 208.1 | 51.8 | 69 | 5.9 | 3.2 | 324.7 | 55.7 | 69 | 5.9 | 3.2 |
| 2. Overprints | 4 | 180.1 | 63.7 | 23 | 19.6 | 9.4 | 343.6 | 24.4 | 27 | 17.9 | 8.6 |
| Section 1, North Limb | | | | | | | | | | | |
| 3. Silverquick conglomerate | 5 | 335.9 | -19.9 | 48 | 11.2 | 5.6 | 322.9 | 57.2 | 30 | 14.1 | 7.1 |
| 4. Overprints and sills | 7 | 340.3 | 51.5 | 115 | 5.6 | 3.0 | 179.9 | 47.9 | 105 | 5.9 | 3.1 |
| Section 2, North Limb | | | | | | | | | | | |
| 5. Powell Creek volcanics | 4 | 299.2 | -13.0 | 74 | 10.8 | 5.0 | 285.2 | 52.8 | 74 | 10.8 | 5.0 |
| Combined | | | | | | | | | | | |
| 6. Silverquick cglm (rows 1 + 3) | 15 | 256.2 | 51.2 | 2 | 38.7 | 21.3 | 324.1 | 56.2 | 53 | 5.3 | 2.9 |
| 7. Silverquick cglm (row 6 + sites 62 and 29) | 17 | 269.9 | 53.7 | 2 | 35.7 | 19.9 | 322.7 | 56.6 | 46 | 5.3 | 3.0 |
| 8. Silverquick + Powell Creek units (rows 5 + 7) | 21 | 282.3 | 35.4 | 2 | 32.0 | 17.9 | 315.2 | 56.8 | 32 | 5.6 | 3.1 |

$N$ is the number of collecting sites, $D_h$, $I_h$ and $D_b$, $I_b$ are declination and inclination of mean directions, relative to present horizontal and relative to bedding calculated using conventional Fisher's statistics, $k$ is Fisher's precision, and $\alpha_{95}$ and $\alpha_{63}$ the radius of the 95% and 63% (standard error) circles of confidence. Bedding attitudes from adjacent sites were averaged together and the mean correction applied to the panel. Mean panel attitudes (bedding dip, bedding downdip azimuth) are as follows: sites 2 to 10 (64° at 358°); sites 11 to 14 (91° at 348°); sites 15 and 16 (89° at 173°); sites 19 to 23 (76° at 171°); sites 50 to 53 (72° at 140°). Combinations are as follows: (1) sites 2 to 10; (2) sites 11 to 14; (3) sites 15, 16, 20, 22, and 23; (4) sites 17, 18 (inverted), 21 (inverted), 24, 25, 26, and 27; and (5) sites 50 to 53.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

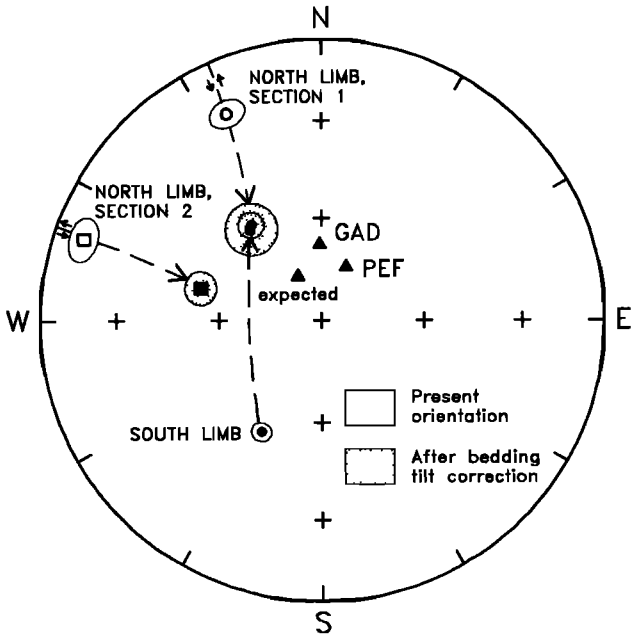



**Figure 14.** Mean directions of the three main sections of Tatlow syncline before and after tilt corrections (rows 1, 3, and 5 and footnotes to Table 2). Standard errors ($\alpha_{63}$) are plotted. See caption to Figure 13 for other symbols.

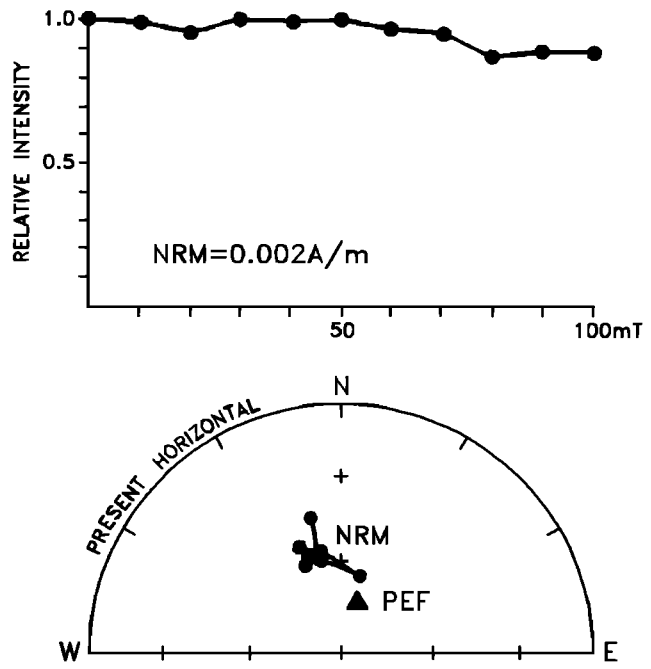



**Figure 16.** AF demagnetization of concretion in fine-grained sandstone, site 24, Silverquick conglomerate. (top) Changes in intensity. (middle) Changes in direction. Symbols and conventions as in Figure 4.

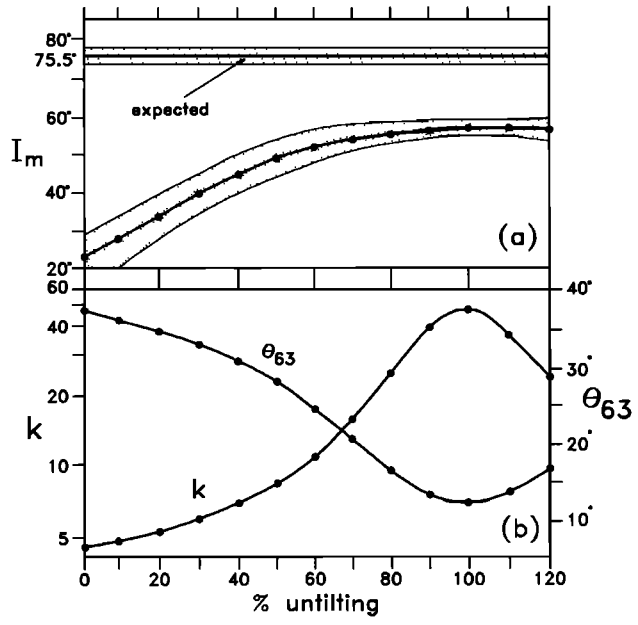



**Figure 15.** (a) Mean inclination ($I_m$), and (b) precision ($k$) and circular standard deviation ($\Theta_{63}$) versus untilting. In Figure 15a, standard error ($\alpha_{63}$) envelopes are plotted. Expected is mean inclination expected for mid-Cretaceous time [*Van Fossen and Kent*, 1992] is plotted. In Figure 15b, block rotation Fisher analysis of data from 21 sites is shown. Untilting of 0% corresponds to beds in their present positions and 100% corresponds to beds fully restored to paleohorizontal using panel tilts given in footnotes to Table 2.

*Wilson et al.* [1968]). Pyrite is rare. The diorites have normal and reversed polarities of magnetization, low coercivities (Figure 18), and $T_{UB}$ between 500° and 550°C and single-component magnetizations, which are probably carried by exsolved magnetite and/or fine-grained disseminated magnetite.

Within each limb of the fold, low $T_{UB}$ magnetizations in bedded rocks share common directions with high $T_{UB}$ magnetizations in diorite intrusions (Table 1). The mean

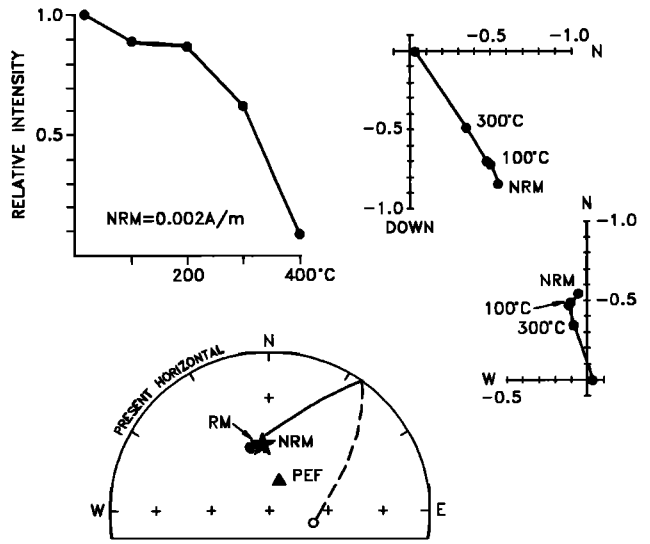



**Figure 17.** Thermal demagnetization of concretion in fine-grained sandstone, site 24. (top left) Changes in intensity. (top right) Normalized orthogonal plots. (bottom) Changes in direction. Symbols and conventions as in Figure 4.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

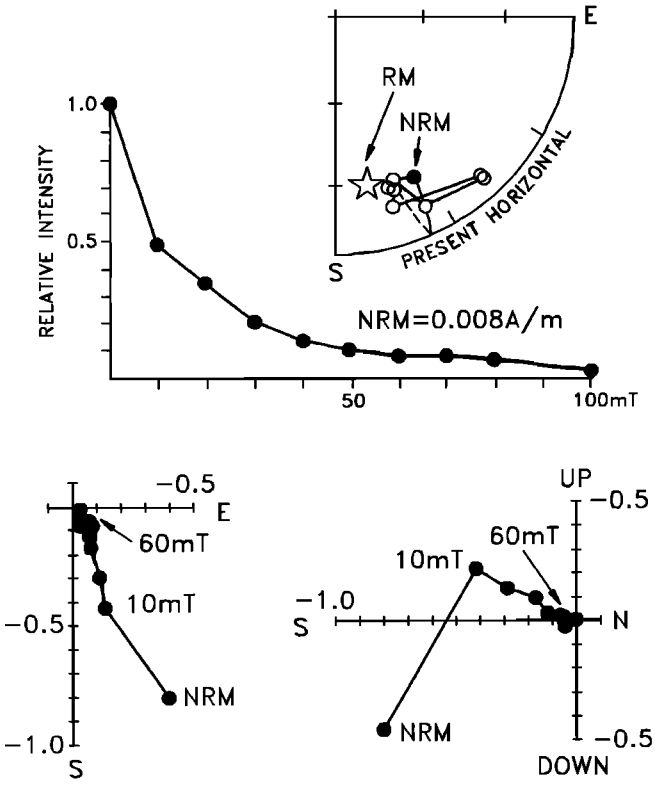



**Figure 18.** AF demagnetization of diorite sill, site 18. (top) Changes in direction, - last two treatment steps have scattered directions and are not plotted. (middle) Changes in intensity. (bottom) Normalized orthogonal plots. RM (star) direction of magnetization removed between 10 and 60 mT. Symbols and conventions as in Figure 4.

directions for each limb (rows 2 and 4, Table 2) describe paths during untilting that overlap and bypass one another (Figure 19). Thus, the magnetization of these 11 sites appears to have been acquired during folding.

On the basis of these observations we propose that the diorite bodies were intruded during folding and that their magnetizations are TRM acquired as they cooled. The magnetizations in the sedimentary rocks and the volcanic breccia sampled are, we propose, syndeformational overprints (CRM) carried by pyrrhotite, which are probably products of the alteration that accompanied intrusion of the diorite bodies. Based on the petrologic and magnetic similarities between the dated dike (site 54) and the other diorite bodies and the presence of both normal and reversed polarities, the age of cooling and magnetization and hence of folding could be Maastrichtian, but this argument is not strong.

## Estimate of Displacement

The calculations in this and following sections have been made using the procedures given by *Beck* [1980] and *Demarest* [1983] and the most recent mid-Cretaceous cratonic reference paleopole 71.2°N, 194.1°E, $A_{95}$ = 3.7, $K$ = 421 [*Van Fossen and Kent,* 1992]. This is based on detailed studies of five rock units spread widely across cratonic North America and spanning the interval 124 to 88 Ma. Errors quoted throughout are for $P$ = 0.05.

Assuming that there has been no relative motion between the sampling area and the craton, the inclination calculated for the Silverquick and Powell Creek strata should be 75.5° ± 2.1° using the above reference paleopole. The observed mean inclination of the prefolding magnetizations is 55.4° ± 3.4° (paleolatitude = 35.9° ± 2.5°). The difference of 20.1° ± 4.0° corresponds to a relative paleolatitudinal displacement (RPD) of the sampling area, northward with respect to the craton, of 26.7° ± 4.3° or 3000 ± 500 km (Figure 20). Because the directions at site 29, north of the Yalakom fault, also fall within the distribution of pretilting magnetizations (Figure 13b), the boundary along which most of this motion occurred probably was east or northeast of site 29. Only a smaller displacement (less than the error of 500 km) could have occurred along the Yalakom Fault (Figure 2). Total displacement on the Yalakom Fault was approximately 115 km [*Kleinspehn,* 1985; *Riddell et al.,* 1993a] less than 50 km of which would have been paleolatitudinal displacement, because the fault strikes WNW/ESE. For the Silverquick beds of the south limb and section 1 of the north limb of the Tatlow syncline, rotations relative to the craton are insignificant (8 ± 11° anticlockwise).

## Discussion

The displacement we infer is in excellent agreement with that estimated for the Mount Stuart Batholith situated farther

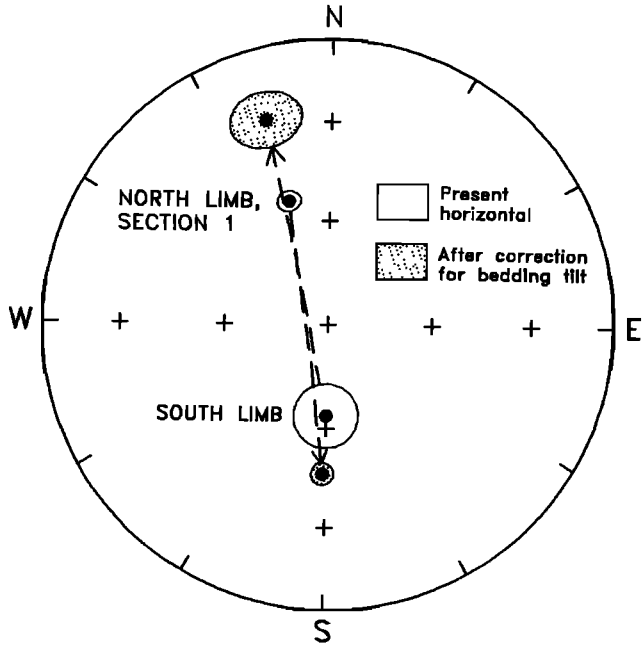



**Figure 19.** Mean direction of magnetizations of diorite intrusions (three sites) and those sites from the Silverquick (seven sites) and Powell Creek (one breccia site) units with low $T_{UB}$ magnetization (sites with asterisks, Table 1). These mean directions correspond to rows 2 and 4 in Table 2. Sites used to obtain mean directions are marked by asterisks in Table 1 and are as follows: South limb, sites 11, 12, 13, and 14; North limb, section 1, sites 17, 18, 21, 24, 25, 26, and 27. Panel dips used are given in Table 2 footnotes. Standard error ($\alpha_{63}$) error circles are plotted. The small circle paths from uncorrected to fully corrected directions intersect before untilting is complete, indicating that the magnetization was acquired during folding.

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

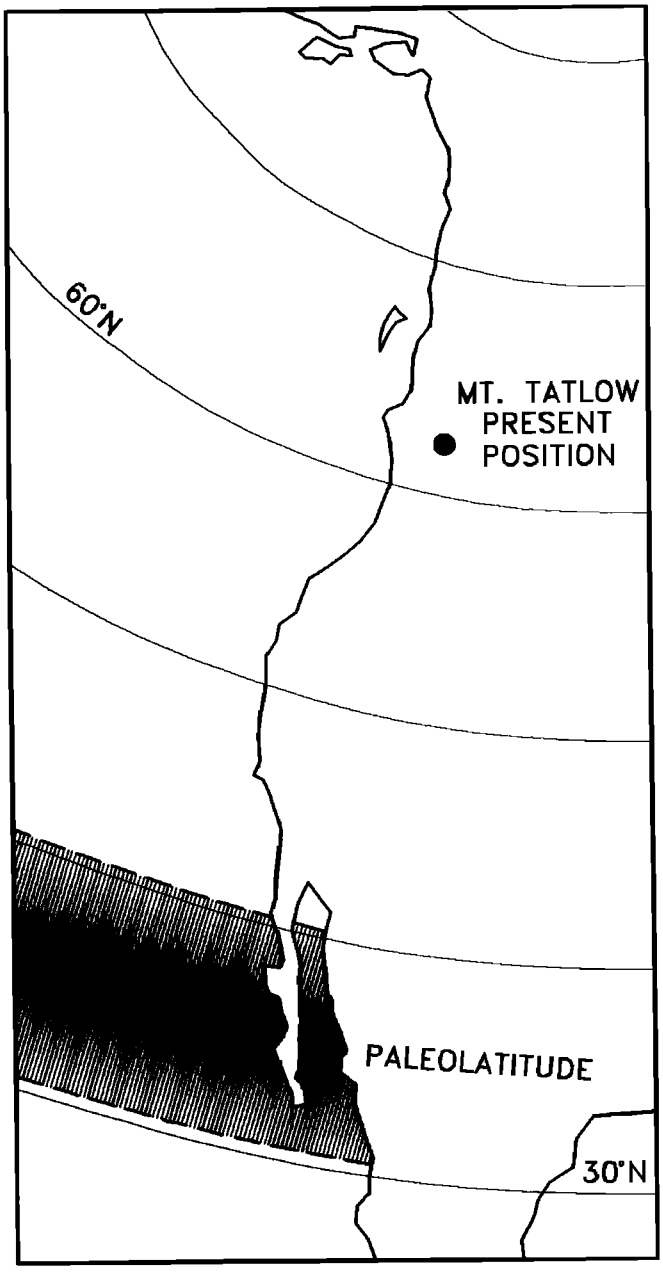



**Figure 20.** Paleolatitude of the Tatlow syncline plotted on a map of western North America with mid-Cretaceous lines of latitude. Observed paleolatitude (35.9°) is 26.7° farther south than that expected (62.6°N). Map is drawn using the mid-Cretaceous cratonic reference paleopole of *Van Fossen and Kent* [1992].

south in the Coast Belt (96 to 93 Ma, MS, Figure 1). In the original study, a RPD from the south of 3500 ± 500 km was estimated assuming the batholith had not been tilted [*Beck and Noson*, 1972; *Beck et al.*, 1981]. *Ague and Brandon* [1992] used hornblende geobarometry to determine that tilt of the sampled outcrops was 7° down toward 145°E, which slightly steepens the mean inclination and reduces the RPD to 3100 ± 600 km (displacement recalculated using *Van Fossen and Kent's* [1992] reference paleopole). The agreement of the Mount Stuart displacement estimate with our estimate (3000 ± 500 km), using an entirely different principle for determining paleohorizontal, is too remarkable to be fortuitous.

For the Spuzzum pluton of the Coast Belt (SP, Figure 1), a displacement of 2500 ± 700 km has been proposed [*Irving et al.*, 1985], based on the assumption of no tilt since emplacement. Using petrologic data, *Brown and Burmester* [1991] argued that the pluton had been tilted first down to the northeast and then to the southwest, so that its present attitude is close to its original one. They proposed that there has been an initial metamorphism in part coincident in time and space with intrusion of the Spuzzum pluton (≈94 Ma), and a second metamorphism spatially associated with the Scuzzy pluton (≈84 Ma), situated to the northeast [*Brown and Burmester*, 1991]. They further propose that the magnetization of Spuzzum dates not from its intrusion but from the second phase of metamorphism and therefore has been affected by only the second phase of tilting, which is sufficient to explain the anomalous Spuzzum direction of magnetization without invoking lateral transport. In contrast, based on a review of numerous age data, *Irving et al.* [1985] argued that the magnetization (observed mainly across the southern part of the Spuzzum pluton and distant from the Scuzzy pluton) was acquired following intrusion between "105 and 90 Ma", and thus would have been affected by both phases of the supposed tilting, which would cancel each other. If they are correct, then the assumption of no tilt, made to obtain the displacement estimate may be essentially correct. Further studies are required to evaluate satisfactorily these competing ideas.

Several other results are available from mid-Cretaceous plutons in the Coast Belt, from farther north around Prince Rupert (PR, Figure 1 inset; 102 and 109 Ma, [*Symons*, 1974, 1977]), and from the south (ST, Figure 1; 102 to 88 Ma [*Vandall*, 1993]); there are no tilt estimates for these bodies, and it is not possible, therefore, to make reliable estimates of translation from them.

Results are also available from the Ventura Member of the Cenomanian to Turonian Midnight Peak Formation (VM, Figure 1), a redbed sequence in the Washington portion of the Tyaughton-Methow Basin [*Burmester et al.*, 1990; *Bazard et al.*, 1990]. These rocks are coeval with the lower Powell Creek volcanics. Magnetization directions are much less scattered when corrected for bedding tilt, but they fall into two distinct groups. *Bazard et al.* [1990] argued that although the data within each group pass a fold test, the magnetizations were not acquired when the beds were lying flat; magnetizations were acquired after the strata were tilted to some degree, but before formation of complex structures observed today; the relevance of this study to the translation problem is questionable [*Bazard et al.*, 1990]. That being said, however, both magnetizations have shallower inclinations than expected, consistent with large displacements from the south that we find.

In summary, our study yields two main results. First, paleomagnetic data from the Coast Belt for which reliable tilt corrections can be made are consistent with the hypothesis that the outer elements (Insular and Coast belts) of the Cordillera have undergone, about 3000 km of northward motion relative to cratonic North America since mid-Cretaceous time [*Beck et al.*, 1981, 1991; *Irving et al.*, 1985; *Ague and Brandon*, 1992; *Umhoefer*, 1987]. Our data flatly contradict models that seek to explain the aberrant Coast

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Belt data by tilting [*Symons*, 1974, 1977; *Butler et al.*, 1989; *Vandall and Palmer*, 1990; *Vandall*, 1993].

Secondly, our result documents very large relative paleolatitudinal motion between the Coast Belt and the adjacent Intermontane Belt (Figure 1, inset). Paleomagnetic results from the mid-Cretaceous Spences Bridge Group (104 Ma; SB Figure 1,) suggest post-mid-Cretaceous northward translation of only 1100 ± 600 km [*Irving and Thorkelson*, 1990; *Irving et al.*, 1995]. Consequently, in order for these belts to achieve their present relative positions, the boundary between the eastern Coast Belt and the Intermontane Belt needs to have accommodated approximately 2000 km of dextral relative motion in Late Cretaceous to Tertiary time.

These two paleomagnetic results, flatly contradict previous geologically based tectonic interpretations of the Canadian cordillera [e.g., *Price and Carmichael*, 1986; *Monger et al.*, 1982, 1994; *Wernicke and Klepaki*, 1988; *van der Heyden*, 1992]. We believe they provide important pieces of the Cordilleran tectonic puzzle. Hopefully, they will provoke critical reviews of existing geological constraints, reexamination of fault displacements, and innovative discussions of Cordilleran evolution in the mobilistic terms demanded by modern plate tectonics.

**Acknowledgments.** Financial support was provided by the Geological Survey of Canada, the Graduate School of the University of Minnesota, Sigma Xi, and National Science Foundation Grants EAR-8804837 and EAR-9005395. Helicopter support and gracious hospitality from White Saddle Air Services and Jen King are gratefully acknowledged. J.A.M. and K.L.K. especially wish to thank Glenn and Joy Woodsworth, who contributed much geological insight, logistical support and fine hospitality. $^{40}Ar/^{39}Ar$ age determinations were made in the laboratory of Tullis Onstott at Princeton University and J.A.M. thanks him for his help. P.J.W. thanks Ian Webster and Larry Diakow at the Geological Survey Branch of British Columbia for help with photomicrographs. We are grateful to R. J. Enkin, Subir Banerjee, Kari Bassett, Mark Brandon, Ralph Haugerud, Rob Coe, and Myrl Beck Jr. for critically reading the manuscript. Their comments led to many improvements. Geological Survey of Canada contribution 39394. Lithoprobe contribution number 640.

## References

Ague, J.J., and M.T. Brandon, Tilt and northward offset of Cordilleran batholiths resolved using igneous barometry, *Nature, 360,* 146-149, 1992.

Bazard, D.R., R.F. Burmester, M.E Beck Jr., J.L. Granirer, and C.G. Schwarz, Paleomagnetism of the Methow region, north-central Washington: Structural application of paleomagnetic data in a complexly deformed, variably remagnetized terrane, *Can. J. Earth Sci., 27,* 330-343, 1990.

Beck, M.E., Jr., Discordant paleomagnetic pole positions as evidence of regional shear in the Western Cordillera of North America, *Am. J. Sci., 266,* 94-712, 1976.

Beck, M.E., Jr., Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. Geophys. Res., 85,* 7115-131, 1980.

Beck, M.E., Jr., Case for northward transport of Baja and coastal southern California: Paleomagnetic data, analysis and alternatives, *Geology, 19,* 506-509, 1991.

Beck, M.E., Jr., Tectonic significance of paleomagnetic results for the western coterminous United States, in *The Geology of North America*, vol. G3, *The Cordilleran Orogen: Coterminous U.S.*, edited B. Burchfiel et al., pp. 683-697, Geological Society of America, Boulder, Colo., 1992.

Beck, M.E., Jr., and L. Noson, Anomalous paleolatitude in Cretaceous granitic rocks, *Nature, 235,* 11-13, 1972.

Beck, M.E., Jr., R.F. Burmester, and R. Schoonover, Paleomagnetism and tectonics of Cretaceous Mt. Stuart batholith of Washington: Translation or tilt?, *Earth Planet. Sci. Lett., 567,* 336-342, 1981.

Bogue, S.W., D.S. Cowan, and J.I. Garver, Paleomagnetic evidence for poleward transport of Upper Jurassic rocks in the Decatur terrane, San Juan Islands, Washington, *J. Geophys. Res., 94,* 10,415-10,427, 1989.

Brown, E.H., and R.F. Burmester, Metamorphic evidence for tilt of the Spuzzum pluton: Diminished basis for Baja British Columbia concept, *Tectonics, 10,* 978-985, 1991.

Burmester, R.F., D.R. Bazard, and M.E. Beck Jr., Post-folding remagnetization that passes the fold test. *Geophys. J. Int., 102,* 455-463, 1990.

Butler, R.F., G.E. Gehrels, W.C. McClelland, S.R. May, and D. Klepacki, Discordant paleomagnetic poles from the Canadian Coast Plutonic Complex: Regional tilt rather than large-scale displacement?, *Geology, 17,* 691-694, 1989.

Cande, S.C., and D.V. Kent, A new geomagnetic polarity time scale for the Late Cretaceous and Cenozoic, *J. Geophys. Res., 97,* 13,917-13,951, 1992.

Cas, R.A.F., and J.V. Wright, *Volcanic Successions: Modern and Ancient,* 528 pp., Allen and Unwin, Winchester, Mass., 1987.

Church, M., and K. Rood, Catalogue of alluvial river channel regime data, *report, edition 1,* Dep. of Geogr., 99 pp., Univ. of B. C., Vancouver, 1983.

Demarest, H.H., Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. Geophys. Res., 88,* 4321-4328, 1983.

Dunlop, D.J., Hysteresis properties of magnetite and their dependence on particle size: A test of pseudo-single-domain remanence models, *J. Geophys. Res., 91,* 9569-9548, 1986.

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London A, 217,* 295-305, 1953.

Garver, J.I., Basin evolution and source terranes of Albian Cenomanian rocks in the Tyaughton basin, southern British Columbia: Implications for mid-Cretaceous tectonics in the Canadian Cordillera, Ph.D. thesis, 227 pp., Univ. of Wash., Seattle, 1989.

Garver, J.I., Provenance of Albian-Cenomanian rocks of the Methow and Tyaughton basins, Southern British Columbia: A mid-Cretaceous link between North America and the Insular Terrane, *Can. J. Earth Sci., 29,* 1274-1295, 1992.

Garver J.I., and M.T. Brandon, Fission track ages of detrital zircons from Cretaceous strata, southern British Columbia: Implications for the Baja BC hypothesis, *Tectonics, 13,* 401-420, 1994.

Garver, J.I., P. Schiarizza, and R.G. Gaba, Stratigraphy and structure of the Eldorado Mountain area, Chilcotin Ranges, southwestern British Columbia (92O/2; 92J/15), in Geologic Fieldwork 1988, *Pap. 1989-1,* pp. 131-143, B. C. Minist. of Energy, Mines, and Pet. Resour., Victoria, 1989.

Glover, J.K., P. Schiarizza, and J.I. Garver, Geology of the Noaxe Creek map area (92O/02), in Geologic Fieldwork 1987, *Pap. 1988-1,* pp. 105-123, B. C. Minist. of Energy, Mines, and Pet. Resour., Victoria, 1988.

Hall, J.M., and B.E. Fisher, The characteristics and significance of secondary magnetitein a profile through dike component of the Troodos, Cyprus, ophiolite, *Can. J. Earth Sci., 24,* 2141-2159, 1987.

Harrison, C.G.A., Analysis of the magnetic vector in a Single

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

rock specimen, *Geophys. J R. Astron. Soc., 60,* 489-492, 1980.

Hickson, C.J., P. Read, W. Mathews, J.A. Hunt, G. Johansen, and G. Rouse, Revised geological mapping of northeastern Taseko Lakes map area, British Columbia,in Current Research, Part A, *Pap. Geol. Surv. of Can., 91-1A,* 207-217, 1991.

Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terranes, *Can. J. Earth Sci., 16,* 669-694, 1979.

Irving, E., and D.J. Thorkelson, On determining paleo-horizontal and latitudinal shifts: Paleomagnetism of Spences Bridge Group, British Columbia, *J. Geophys. Res., 95,* 19,213-19,234, 1990.

Irving, E., and P.J. Wynne, Paleomagnetic evidence bearing on the evolution of the Canadian Cordillera, *Philos. Trans. R. Soc. London A, 331,* 487-509, 1990.

Irving, E., G.J. Woodsworth, P.J. Wynne, and A. Morrison, Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia, *Can. J. Earth Sci., 22,* 584-598, 1985.

Irving, E., D.J. Thorkelson, P.M. Wheadon, and R.J. Enkin, Paleomagnetism of the Spences Bridge Group and northward displacement of the Intermontane Belt, British Columbia: A second look, *J. Geophys. Res.*, this issue.

Jeletzky, J.A., and H.W. Tipper, Upper Jurassic and Cretaceous rocks of Taseko Lakes map-area and their bearing on the geological history of southwestern British Columbia, *Pap. Geol. Surv. of Can., 67-54,* 1968.

Kent, D.V., and F.M. Gradstein, A Jurassic to recent chronology, in *The Geology of North America*, vol. M, *The Western North Atlantic Region,* edited by P.R. Vogt and B.E. Tucholke, pp. 45-50, Geological Society of America, Boulder, Colo., 1986.

Kleinspehn, K.L., Cretaceous sedimentation and tectonics, Tyaughton-Methow Basin, southwestern British Columbia, *Can. J. Earth Sci., 22,* 154-174, 1985.

McFadden, P.L., and A.B. Reid, Analysis of palaeomagnetic inclination data, *Geophys. J. R. Astron. Soc., 69,* 309-319, 1982.

McLaren, G.P., Geology and lithogeochemistry of the Chilko -Taseko Lakes area (92)/4,5; 92J/13; 92K/16; 92N/1), in Geologic Fieldwork 1986, *Open File Map 1987-12,* B. C. Minist. of Energy, Mines, and Pet. Resour., Victoria, 1987.

Monger, J.W.H., and E. Irving, Northward displacement of north-central British Columbia, *Nature, 285,* 289-294, 1980.

Monger, J.W.H., R.A. Price, and D.J. Templeman-Kluit, Tectonic accretion and the origin of the two major metamorphic welt in the Canadian Cordillera, *Geology, 10,* 70-75, 1982.

Monger, J.W.H., P. van der Heyden, J.M. Journeay, C.A. Evanchik, and J.B. Mahoney, Jurassic-Cretaceous basins along the Canadian Coast Belt: Their bearing on pre-mid-Cretaceous sinistral displacements, *Geology, 22,* 175-178, 1994.

Price, R.A., and D.M. Carmichael, Geometric test for Late Cretaceous-Paleogene intracontinental transform faulting in the Canadian Cordillera, *Geology, 14,* 468-471, 1986.

Riddell, J., P. Schiarrizza, R.G. Gaba, N. Caira, and A. Findlay, Geology and mineral occurrences of the Mount Tatlow area (92O/5,6 and 12), in Geological Fieldwork 1992, *Pap. 1993-1,* pp. 37-52, B. C. Minist. of Energy, Mines, and Pet. Resour., Victoria, 1993a.

Riddell, J., P. Schiarrizza, R. Gaba, G. McLaren, and J. Rouse, Geology of the Mount Tatlow map area, (920/5, 6, 12), British Columbia, *Open File Map 1993-8,* B. C. Minist. of Energy, Mines, and Pet. Resour., Victoria, 1993b.

Roberts, R.A., and G.M. Turner, Diagenetic formation of ferrimagnetic iron sulphide minerals in rapidly deposited marine sediments, South Island, New Zealand, *Earth Planet. Sci. Lett., 115,* 257-273, 1993.

Rusmore, M.E., and G.J. Woodsworth, Coast Plutonic Com plex: A mid-Cretaceous contractional orogen, *Geology, 19,* 941-944, 1991.

Symons, D.T.A., Age and tectonic implications of paleo magnetic results from plutons near Prince Rupert, *J. Geophys. Res., 79,* 2690-2697, 1974.

Symons, D.T.A., Paleomagnetism of Mesozoic plutons in the westernmost Coast Complex of British Columbia, *Can. J. Earth Sci. 14,* 2127-2139, 1977.

Tipper, H.W., Mesozoic and Cenozoic geology of the northeast part of Mount Waddington map-area (92N) Coast District, British Columbia, *Pap. Geol. Surv. of Can., 68-33,* 1969.

Tipper, H.W., Taseko Lakes (92O) map area, *Open File 534,* Geol. Surv. of Can., Ottawa, 1978.

Umhoefer, P.J., Northward translation of "Baja British Columbia" along the Late Cretaceous to Paleocene margin of western North America, *Tectonics, 6,* 377-395, 1987.

Umhoefer, P.J., and K.L. Kleinspehn, Mesoscale and regional kinematics of the northwestern Yalakom fault system: Major Paleogene dextral faulting in British Columbia, Canada, *Tectonics,* in press, 1995.

Vandall, T.A., Cretaceous Coast Belt paleomagnetic data from the Spetch Creek pluton B.C.: Evidence for tilt and moderate displacement model, *Can. J. Earth Sci., 29,* 1037-1048, 1993.

Vandall, T.A., and H.C. Palmer, Upper limit of docking time for Stikinia and Terrane I: Paleomagnetic evidence from the Eocene Ootsa Lake Group, British Columbia, *Can. J. Earth Sci., 27,* 212-218, 1990.

van der Heyden, P., A Middle Jurassic to early Tertiary Andean-Sierran arc model for the Coast Belt of British Columbia, *Tectonics, 11,* 82-97, 1992.

Van Fossen, M.C., and D.V. Kent, Paleomagnetism of 122 Ma plutons in New England and the Mid-Cretaceous paleomagnetic field in North America: True polar wander or large-scale differential mantle motion?, *J. Geophys. Res., 97,* 19,651-19,661, 1992.

Wernicke, B.P., and D.W. Klepacki, Escape hypothesis for the Stikine block, *Geology 16,* 461-464, 1988.

Wheeler, J.O., and P. McFeely (Compilors), Tectonic assemblage map of Canadian Cordillera and adjacent points of United States of America, *Map 1712A,* Geol. Surv. of Can., Vancouver, 1991.

Wilson, R.L., S.E. Haggerty, and N.D. Watkins, Variation of paleomagnetic stability and other parameters in a vertical traverse of a single Icelandic lava, *Geophys. J. R. Astron. Soc., 16,* 79-96, 1968.

E. Irving and P.J. Wynne, Pacific Geoscience Centre, Box 6000, Sidney, B.C., Canada V8L 4B2. (e-mail: wynne@pgc.emr.ca)

K.L. Kleinspehn, Department of Geology and Geophysics, University of Minnesota, 310 Pilsbury Drive SE, Minneapolis, MN 55455. (e-mail: klein004@maroon.tc.umn.edu)

J.A. Maxson, Department of Geology, Carleton College, Northfield, MN 55057. (e-mail: jmaxson@carleton.edu)

(Received May 11, 1994; revised September 28, 1994; accepted October 6, 1994.)

21562202b, 1995, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/94JB02643 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License